

Machine Learning

- Machine Learning is a **branch of AI** that **enables systems to learn from data** and improve without explicit programming.
- Machine Learning is focused on **building systems capable of learning from data, identifying patterns, and making decisions** with minimal human intervention.
- Instead of being explicitly programmed to perform a task, **ML algorithms are trained on large amounts of data and use that data to make predictions or decisions.**



Figure - Machine Learning Workflow

Why do we use Machine Learning?

- We use Machine Learning(ML) because it **enables computers to automatically learn from data and make decisions or predictions** without being explicitly programmed for every specific task.
- It is ideal for solving complex problems where traditional rule-based approaches fall short.
- Example : **Email Spam Detection**
- Suppose we want to build a system that can detect spam emails.

Traditional programming approach



- Traditional programming relies on human-written rules.
- In traditional programming, a programmer writes explicit rules or instructions for the computer to follow.
- These rules dictate exactly how the computer should process input data to produce the desired output.
- It requires a deep understanding of the problem and a clear way to encode the solution in a programming language.

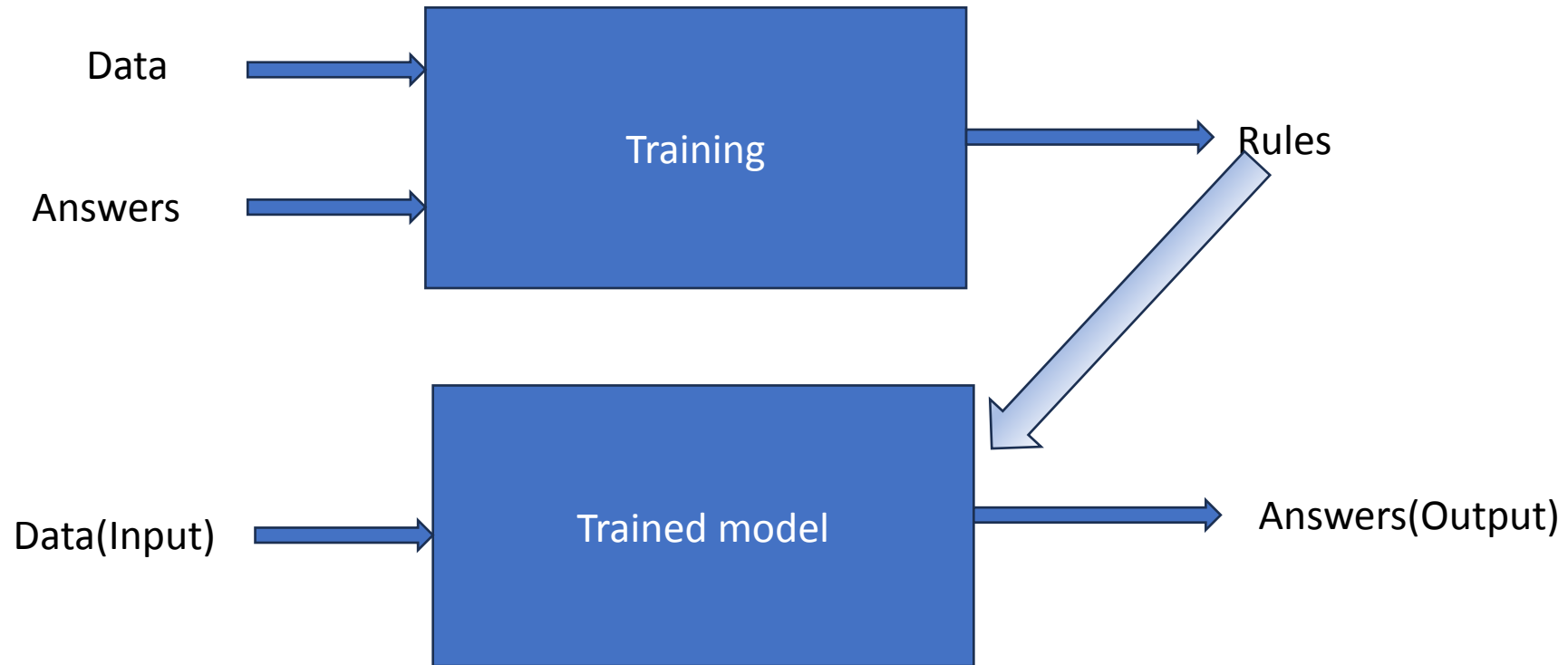
Traditional programming approach

- Traditionally, developers implemented spam filters using fixed rules, explicitly defining what counts as spam.
- You would write a set of specific rules, like
 - Keyword Matching**
 - If the subject contains “**WIN A PRIZE**,” mark it as spam
 - If the body includes “**Click here for free money**,” mark as spam.
 - Too Many Links**
 - If the email contains more than five links, mark as spam.
- But spammers constantly change tactics and hard-coding every new pattern becomes impractical.
- For example:
 - “W!N A PR!ZE” instead of “WIN A PRIZE”
 - Embedding promotional text inside an image instead of the email body

Machine Learning approach

- We can train a model using a dataset of labeled emails(spam and not spam).
- The model learns patterns, such as common spam keywords, send behavior, or formatting styles.
- Once trained, the model can accurately predict whether a new email is spam – even if it has never seen that exact message before.
- Over time, as it is exposed to more data, it can adapt and improve its accuracy.

Machine Learning approach



Machine Learning

- In machine learning, instead of writing explicit rules, a programmer trains a model using a large dataset.
- The model learns patterns and relationships from the data, enabling it to make predictions or decisions without being explicitly programmed for each possibility.
- This approach is particularly useful for complex problems where defining explicit rules is difficult or impossible.

Applications of Machine Learning

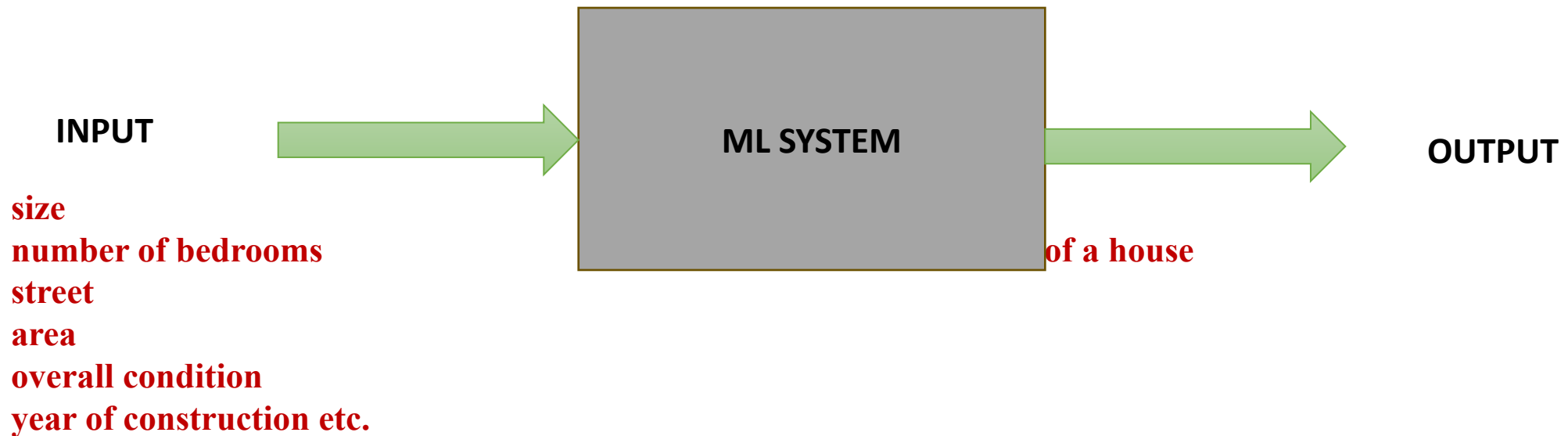
- Product Image Classification on Production Lines
- Tumor Detection in Brain Scans
- News Article Categorization
- Flagging Offensive Content Online
- Automatic text summarization
- Building chatbots or Virtual Assistants
- Voice command recognition
- Product recommendation systems

FEATURES

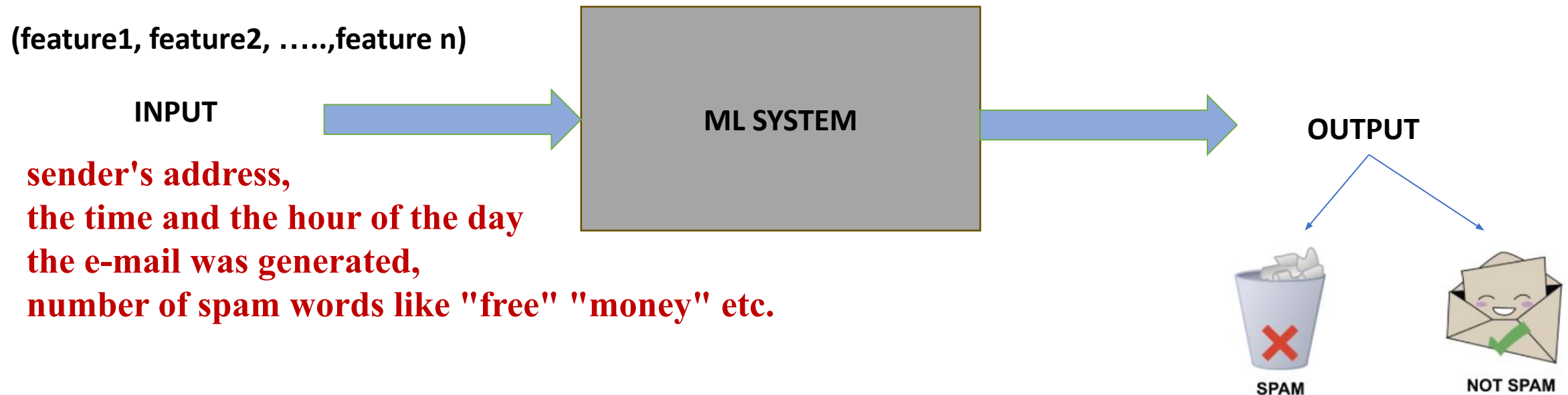
- In machine learning, **features** are the **input data**.
- A single feature represent a specific input variable.
- They are also known as **independent variables, predictors, or attributes**.

Example : Consider the task of **predicting the market price of a house.**
(feature1, feature2,,feature n)

HOUSE PRICE PREDICTION USING MACHINE LEARNING TECHNIQUES



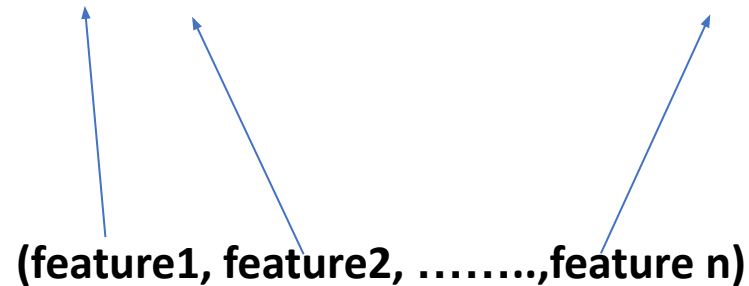
Example : an e-mail spam detector



• Features :

- Mathematically, the **list of input features** will be represented as **a vector** with the size of N like x_1 , x_2 , x_3 until x_n .

$$X = (x_1 , x_2 , x_3 , \dots, x_n)$$

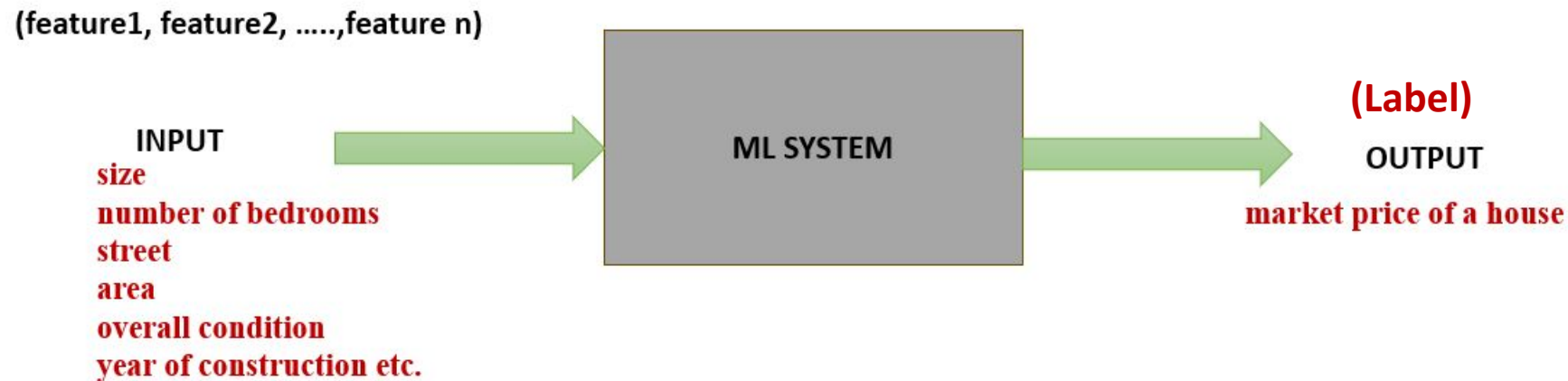


(feature1, feature2,,feature n)

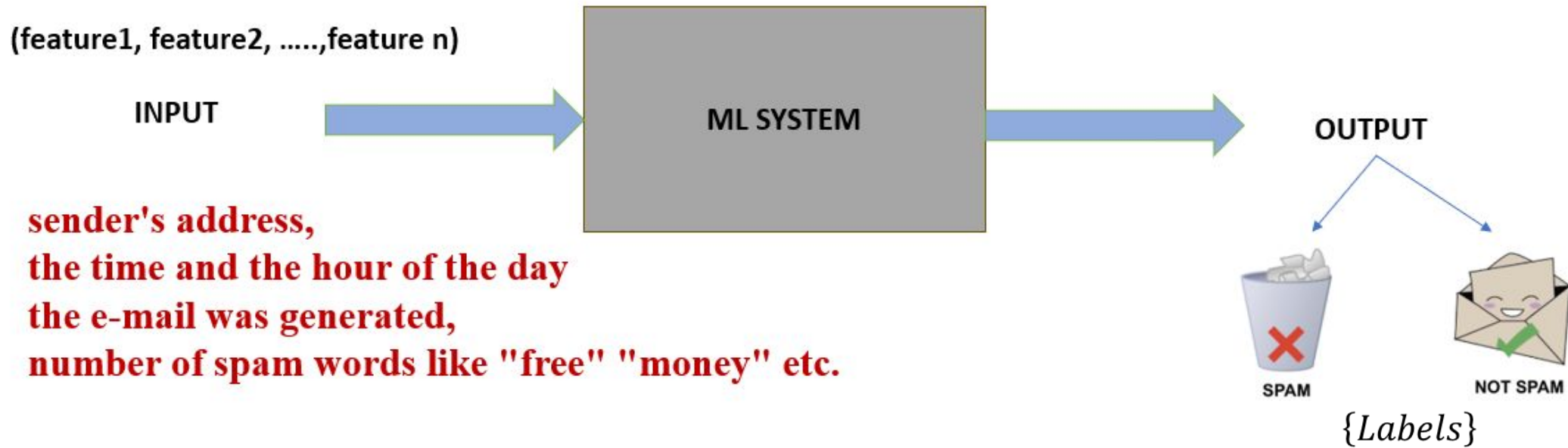
The diagram shows three blue arrows pointing upwards from the text "(feature1, feature2,,feature n)" to the corresponding components of the vector X: from "feature1" to x_1 , from "feature2" to x_2 , and from "feature n" to x_n .

Labels :

- Labels are the **target variables that you want to predict.**
- These are the **output** of your machine learning model, which the algorithm learns to predict based on the features.
- In the housing price example, the **label** would be **the price of the house.**
- Also known as target variable, output, response or dependent variable.



Labels :



Basics of Machine Learning

Features and Labels :

Mathematically,

The diagram illustrates the components of the machine learning equation $Y = f(X)$. The word "Label" is written in red above the variable Y , and the word "Features" is written in red above the variable X . Two blue arrows point from these labels down to their respective variables in the equation.

$$Y = f(X)$$

Y equal to the function of X.

Labeled data :

Labeledexample :
(x , y)
(features , **label**)

- Labeled data is **data that has been assigned a label** or category, indicating the correct classification for each data point.
- This labelling is typically done by human annotators and is crucial for **supervised learning tasks**.
- In supervised learning, the model learns from labelled examples to make predictions on new, unseen data.
- **Examples of labelled data include:**
 - A dataset of images with labels indicating whether each image contains a cat or a dog.
 - An email dataset labelled as spam or not spam.

Unlabeled data :

Unlabeled example :

(x , ?)

(features , ?)

- Unlabeled data is data that **does not have any labels** or categories assigned to it.
- The true classification or category of each data point is unknown, making unlabeled data suitable for **unsupervised learning tasks**.
- In unsupervised learning, the model must learn from the inherent structure of the data to uncover patterns.

Examples of unlabeled data include:

- An image dataset without any labels indicating the objects in each image.
- A dataset of customer transactions without any labels indicating fraudulent or non-fraudulent transactions.

Machine Learning Paradigms

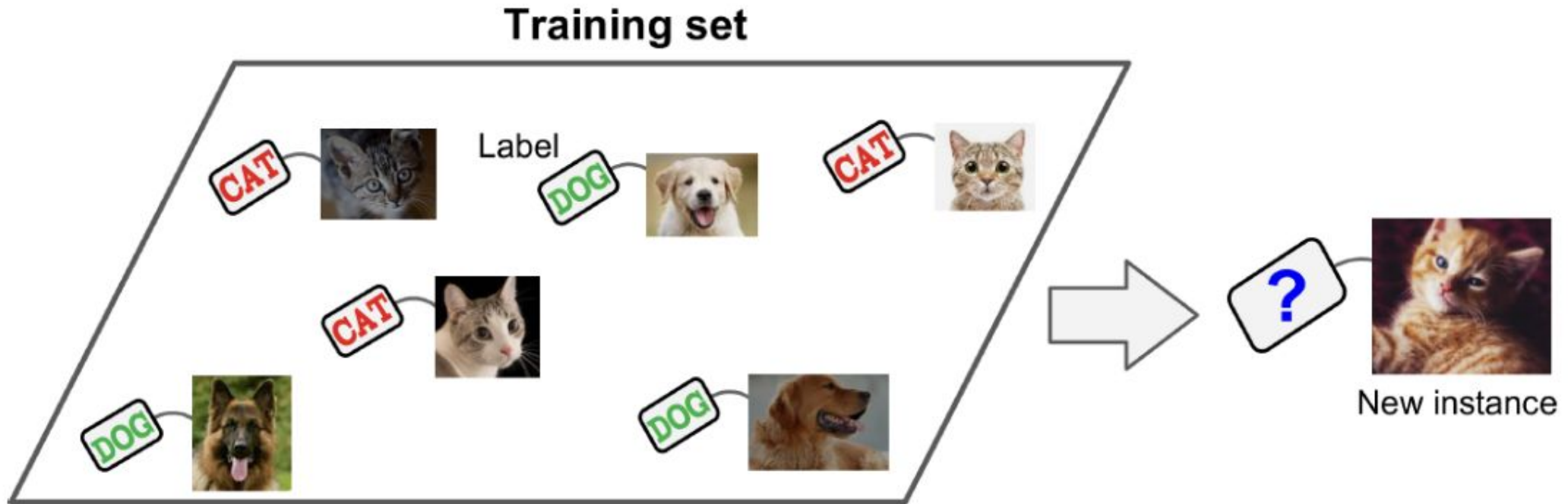
Machine Learning systems can be classified according to the **amount and type of supervision** they get during training.

1. **Supervised Learning**
2. **Unsupervised Learning**
3. **Semi-Supervised Learning**
4. **Reinforcement Learning**

Supervised Learning

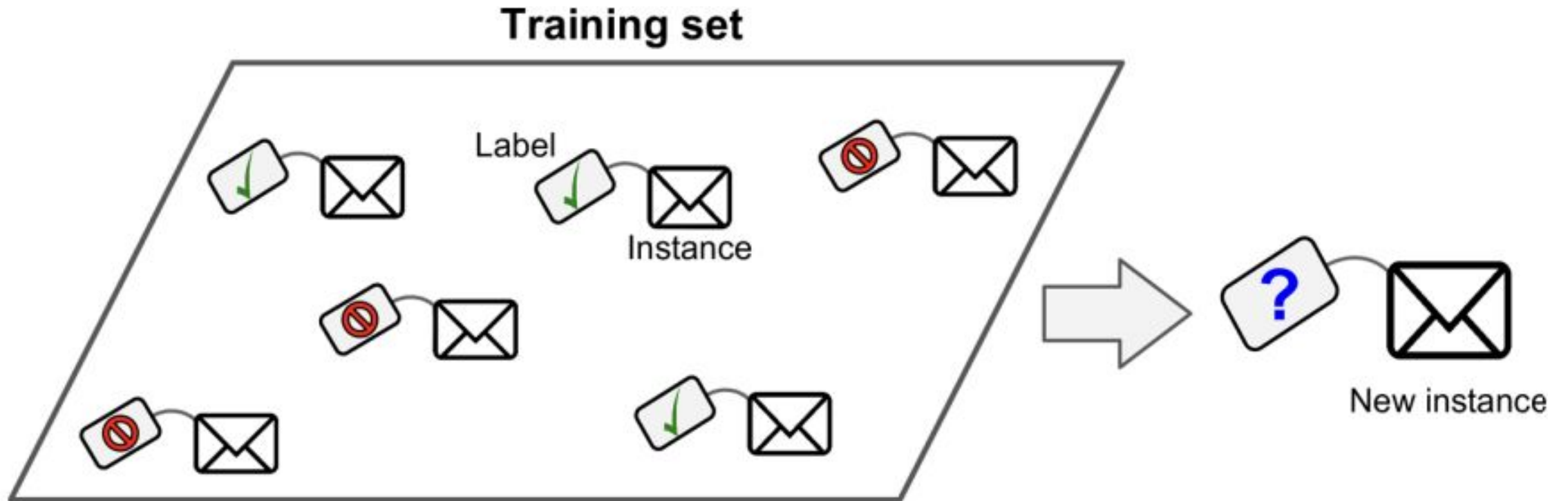
- Supervised learning is the most common type of machine learning.
- In supervised learning, the algorithm is **on a labeled dataset**.
- Each training example includes **an input(features) and an output(label)**.
- Goal for the model is to **learn a mapping function from inputs to outputs**.
- This allows the algorithm to make predictions or decisions when given new, unseen data.
- A **labeled training set** refers to a collection of data used in machine learning where each data point is paired with a corresponding label.
- These labels typically represent the correct output or category that the model should predict for each input.

Supervised Learning



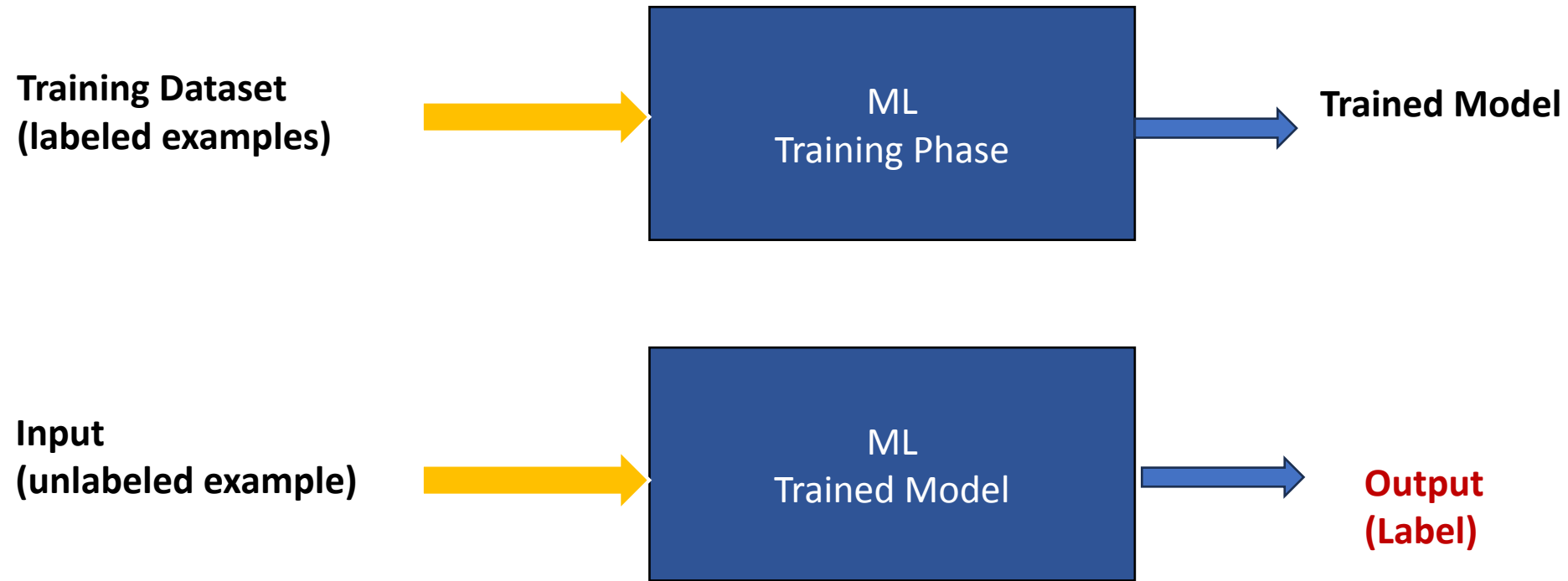
A labeled training set for supervised learning (e.g., Image classification)

Supervised Learning



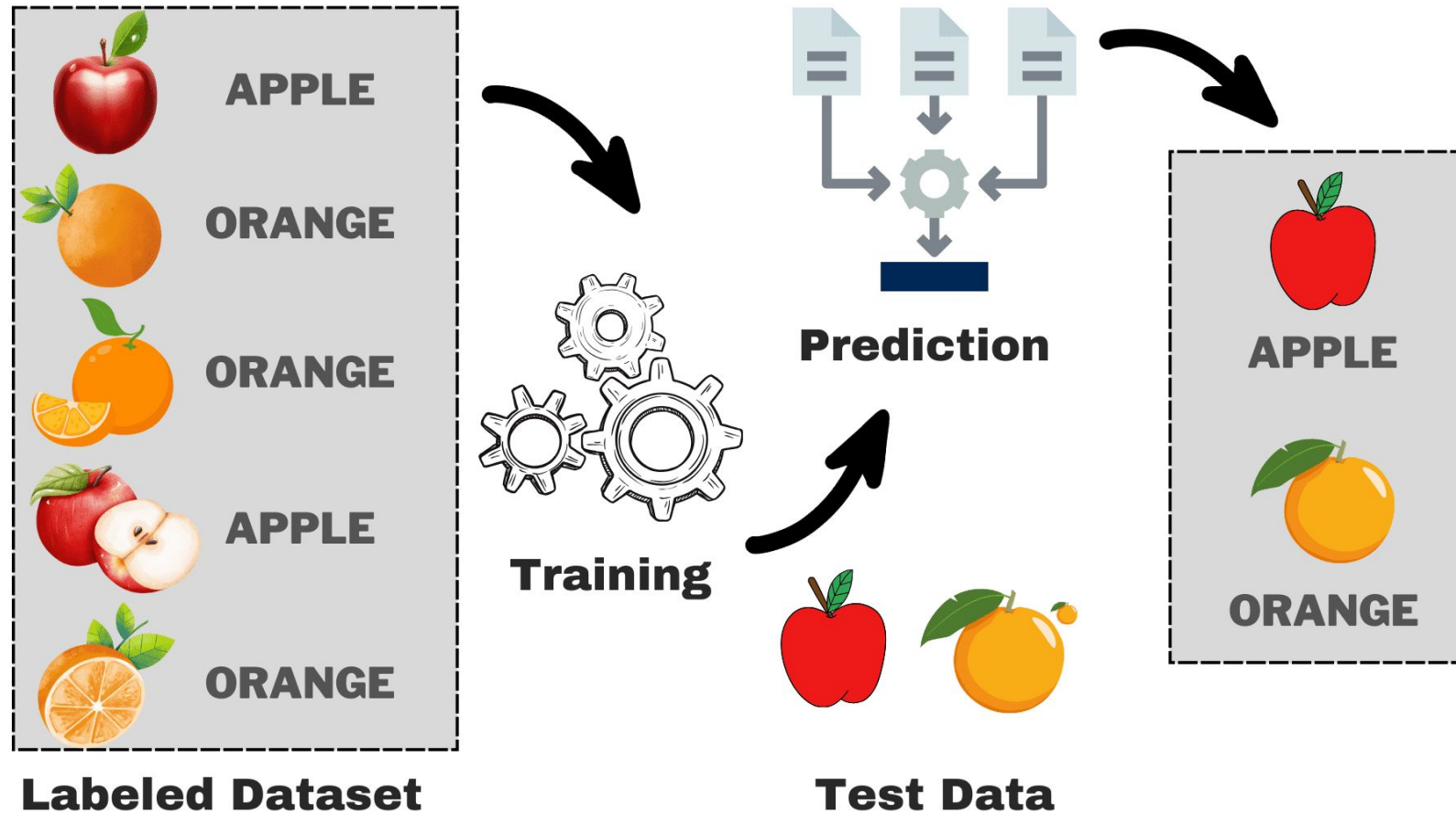
A labeled training set for spam classification (an example of supervised learning)

Supervised Learning



Supervised learning model example

Apple/Orange Classification



Supervised learning model example

House price prediction

S l N o	price	area	be dr oo ms	bathr ooms	stor ies	main road	gustr oom	base ment	hotwater heating	aircondi tioning	parkin g	prefare a	furnishingstatus
1	1330000	7420	4	2	3	yes	no	no	no	yes	2	yes	furnished
2	1225000	8960	4	4	4	yes	no	no	no	yes	3	no	furnished
3	1225000	9960	3	2	2	yes	no	yes	no	no	2	yes	semi-furnished
4	1221500	7500	4	2	2	yes	no	yes	no	yes	3	yes	furnished
5	?	7420	4	1	2	yes	yes	yes	no	yes	2	no	furnished

Supervised Learning

Phases in Supervised Learning :

- **Training Phase:-** **Model learns from the labeled data.**
 - **Input :** Labeled dataset (features + corresponding labels)
 - **Model selection :** Choose a suitable algorithm (eg : Linear Regression, SVM, Decision Tree)
 - **Learning :** Algorithm finds the patterns that map inputs to correct outputs.
 - **Output :** A trained model that has “learned” from the data.
- **Testing Phase:-** **Trained model is evaluated using new, unseen data.**
 - **Input :** Test data(features only)
 - **Prediction(Output) :** Model makes predictions based on what it learned.
 - **Evaluation :** Compare predictions with actual labels using metrics like accuracy, RMSE etc.

Supervised Learning

Supervised learning tasks can be divided into two types:

- 1. Classification**
- 2. Regression.**

Supervised Learning

1. Classification :

- A typical supervised learning task is classification.
- In classification, the **label represents a categorization, or class.**
- Classification models **predict discrete outcomes(discrete labels), or categories.**
- Goal is to **assign an input data to one of several predefined classes.**
- For instance, **classifying emails as spam or non-spam** based on their content is an example of classification.

There are two common classification scenarios.

i. Binary classification

- **It categorizes data points into one of two distinct classes.**
- **Example1:** classifying emails as spam or non-spam.
- **Example2:** In a medical diagnosis, a binary classifier for a specific disease could take a patient's symptoms as input features and predict whether the patient is healthy or has the disease. The possible outcomes of the diagnosis are positive and negative.

- **Binary classification examples :**

Application	Observation	0	1
Medical Diagnosis	Patient	Healthy	Diseased
E-mail Analysis	E-mail	Not Spam	Spam
Financial Data Analysis	Transaction	Not Fraud	Fraud
Marketing	Website visitor	Won't Buy	Will Buy
Image Classification	Image	dog	Not dog

ii. Multiclass classification

- Multiclass classification extends binary classification to **predict a label that represents one of multiple possible classes.**

Examples :

- The **genre of a movie (comedy, horror, romance, adventure, or science fiction)** based on its cast, director, and budget.
- An example of multiclass classification, using **images of vegetables**, where each image is **either a carrot, tomato, or potato.**
- Task of identifying the **color of a specific flower** is **yellow, green, red or blue.**

- Common classification algorithms :
 - **Support Vector Machine(SVM)**
 - **Logistic Regression**
 - **Naive Bayes classifier**
 - **Decision Trees**
 - **Random Forest Classifiers**
 - **K-Nearest Neighbors (KNN).**

Supervised Learning

2. Regression :

- Regression models **predict continuous values(real values)**.
- Regression is a form of supervised machine learning in which the label predicted by the model is a numeric value.

Examples :

- Predicting house prices
- Estimating a person's weight based on height and age
- Forecasting stock prices

Supervised Learning

2. Regression :

- Common regression algorithms :
 - **Linear Regression**
 - **Polynomial Regression**
 - **Ridge Regression**
 - **Support Vector Regression(SVR)**
 - **Decision Tree Regressor**
 - **KNN Regression(K-Nearest Neighbors Regressor)**

Unsupervised Learning

- Unsupervised learning is a type of machine learning where the **model is trained on unlabeled data.**
- Unsupervised learning **is learning without a teacher** supervising the learning process.
- The system tries to identify patterns, structures or grouping in the input data without predefined outputs.
- It is particularly useful when dealing with large datasets where manual labeling would be impractical or costly.
- Useful for exploratory data analysis, data clustering and dimensionality reduction.

Unsupervised Learning

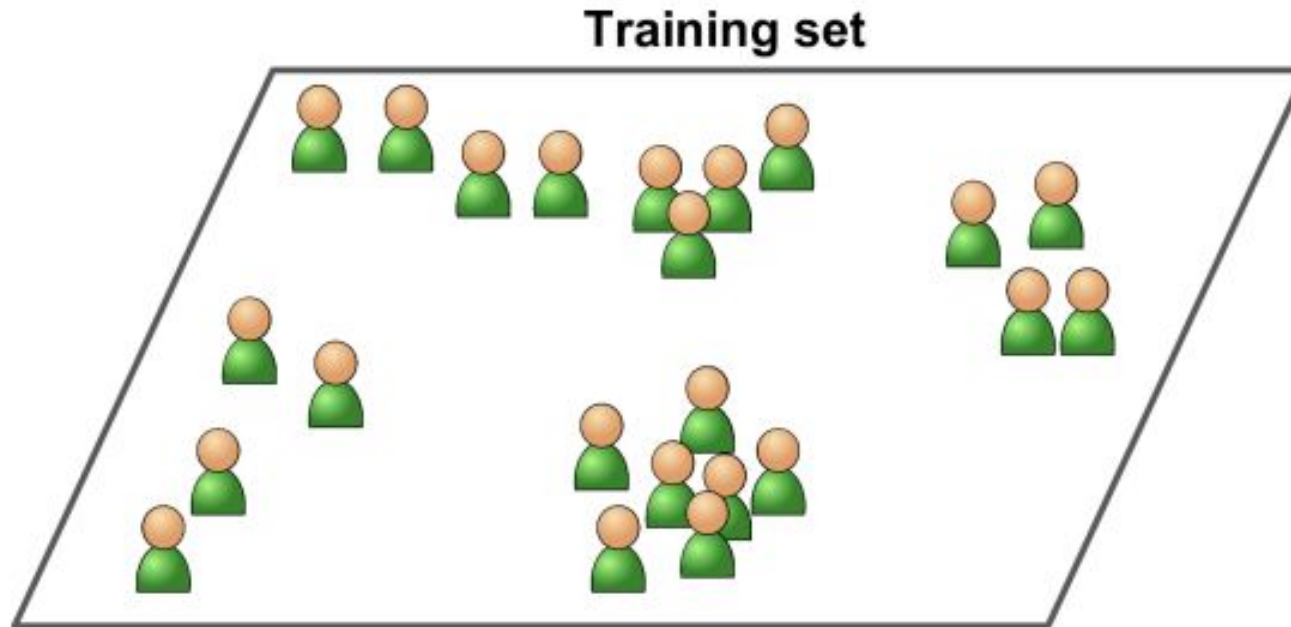
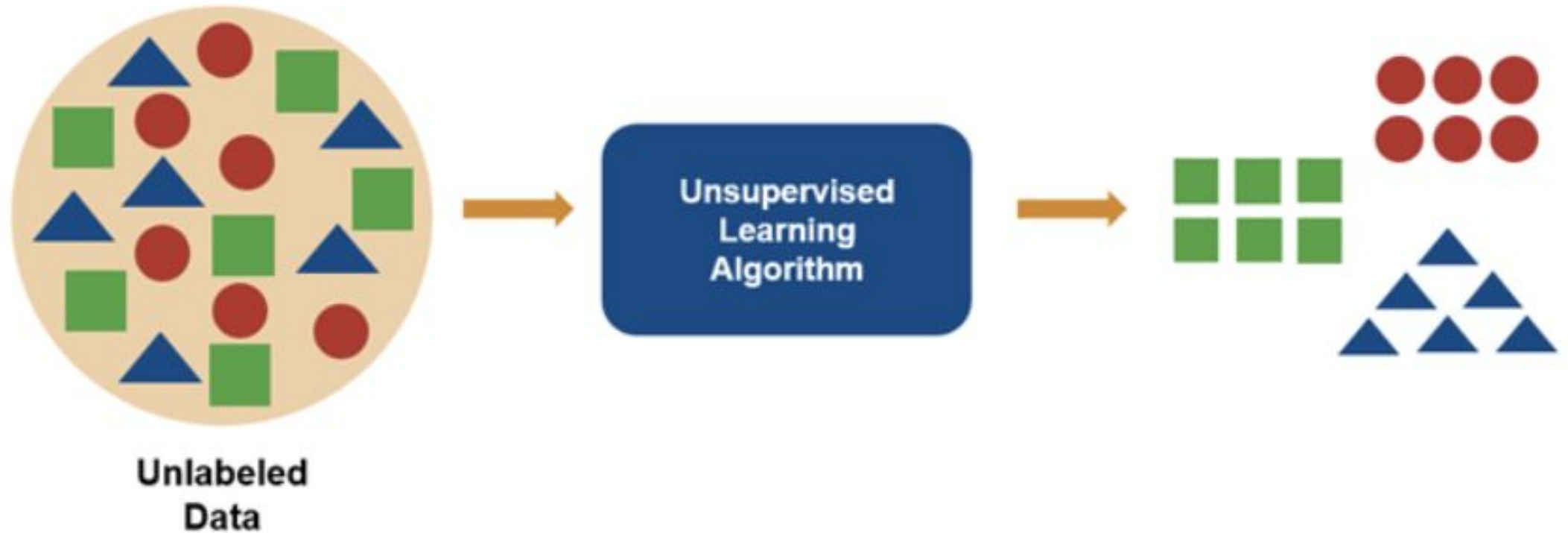


Figure - An unlabeled training set for unsupervised learning

Unsupervised Learning



Organizing unlabeled data into groups using unsupervised learning.

Some of the most important unsupervised learning algorithms :

- Clustering
 - K-Means
 - DBSCAN
 - Hierarchical Cluster Analysis (HCA)
- Anomaly detection and novelty detection
 - One-class SVM
 - Isolation Forest
- Visualization and dimensionality reduction
 - Principal Component Analysis (PCA)
 - Kernel PCA
 - Locally Linear Embedding (LLE)
 - t-Distributed Stochastic Neighbor Embedding (t-SNE)
- Association rule learning
 - Apriori
 - Eclat

Types of Unsupervised Learning Tasks

1. Clustering:

- Clustering is the **most common unsupervised learning method**.
- Used to **group similar data points into clusters** without prior labels.
- If you use a hierarchical clustering algorithm, it may also subdivide each group into smaller groups
- It is used for exploratory data analysis, pattern recognition, anomaly detection, image segmentation, and more.
 - **K-Means**
 - **DBSCAN**(Density-Based Spatial Clustering of Applications with Noise)
 - **Hierarchical Cluster Analysis(HCA)**

Types of Unsupervised Learning Tasks

1. Clustering:

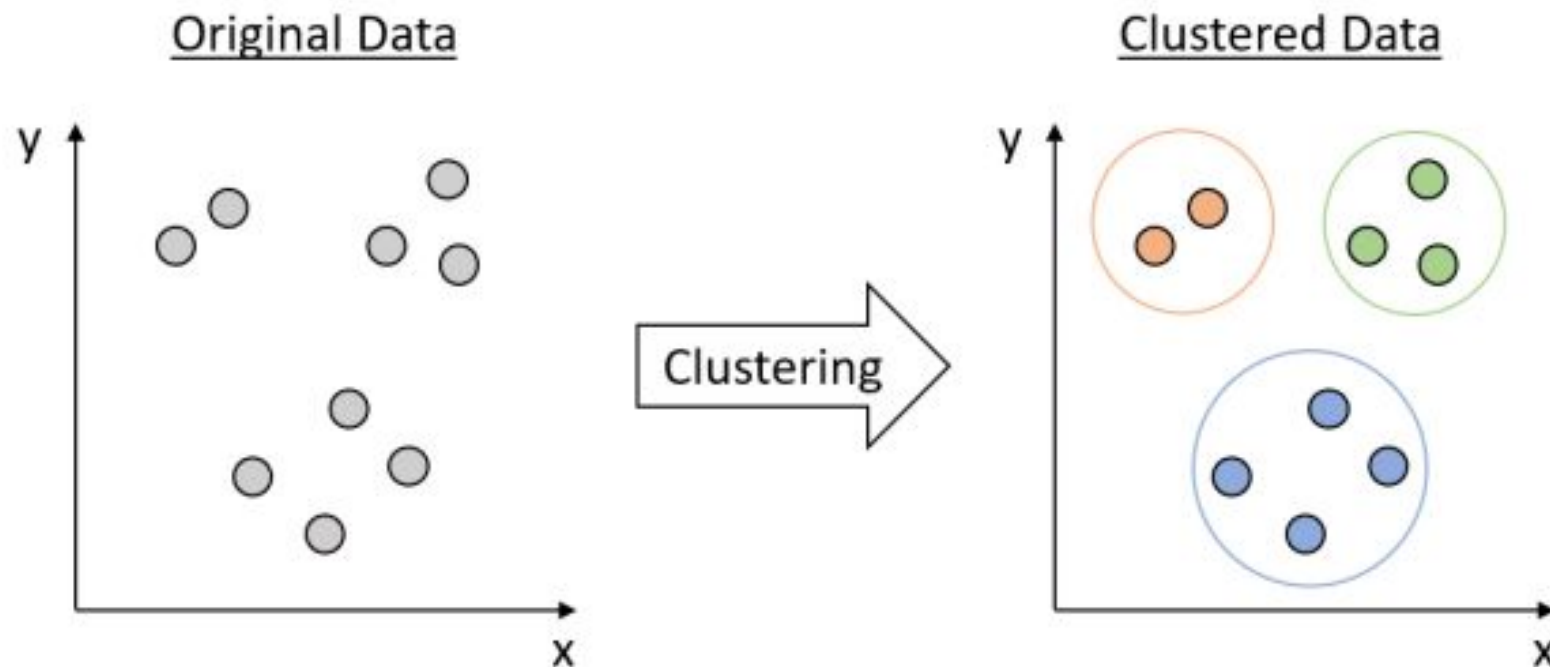
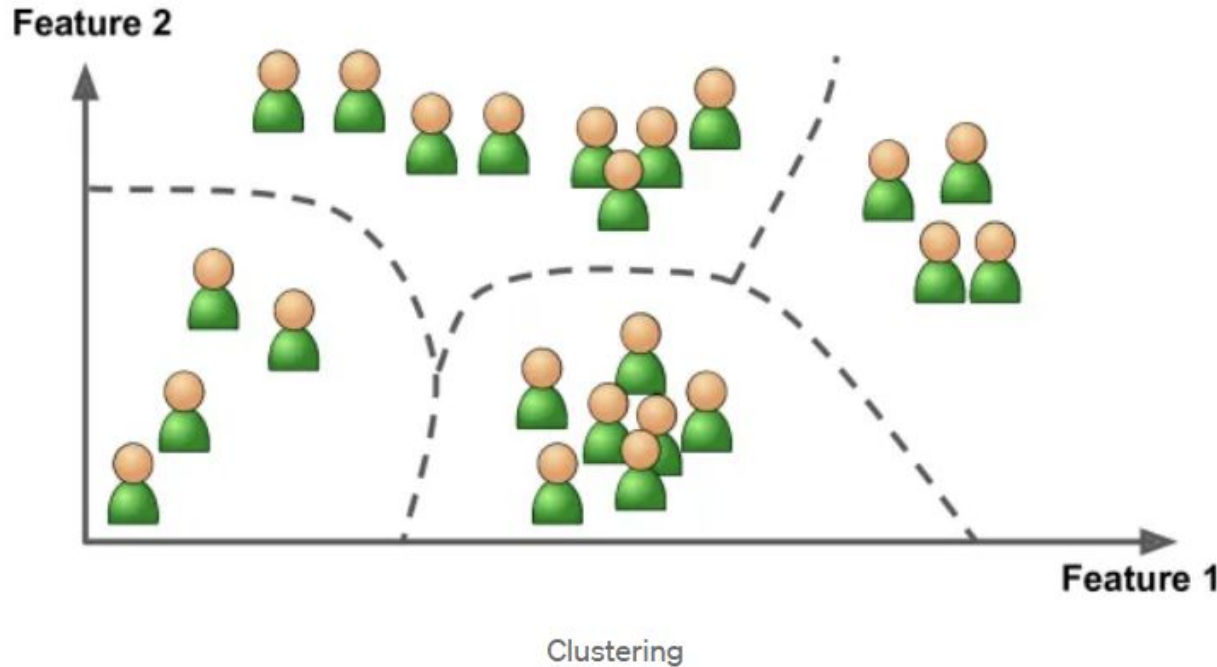


Figure: Graphical representation of how clustering can be used to identify patterns in a data set.

Types of Unsupervised Learning Tasks

1. Clustering:



Say you have a lot of data about your blog's visitors. You may want to run a clustering algorithm to try to detect groups of similar visitors. At no point do you tell the algorithm which group a visitor belongs to: it finds those connections without your help.

For example, it (the algorithm) might notice that 40% of your visitors are males who love comic books and generally read your blog in the evening, while 20% are young sci-fi lovers who visit during the weekends.

If you use a hierarchical clustering algorithm, it may also subdivide each group into smaller groups. This may help you target your posts for each group.

2. Anomaly Detection and Novelty Detection:

Anomaly detection : Used to **identify unusual patterns in data**.

- The system is shown mostly normal instances during training, so it learns to recognize them;
- then, when it sees a new instance, it can tell whether it looks like a normal one or whether it is likely an anomaly.

Examples :

- **detecting unusual credit card transactions to prevent fraud**
- **catching manufacturing defects**
- **automatically removing outliers from a dataset before feeding it to another learning algorithm.**

2. Anomaly Detection and Novelty Detection:

Anomaly detection :



Figure - Anomaly detection

2. Anomaly Detection and Novelty Detection:

Novelty detection : Very similar Anomaly Detection,

- It aims to **detect new instances that look different from all instances in the training set.**
- This requires having a very “**clean**” **training set**, devoid of any instance that you would like the algorithm to detect.
- Difference- if you have thousands of pictures of dogs, and 1% of these pictures represent Chihuahuas then-
- **A novelty detection algorithm should not treat new pictures of Chihuahuas as novelties, i.e it will accept them as normal.**
- On the other hand, anomaly detection algorithms may consider these dogs as so rare (as mentioned 1%) and so different (Refer image) from other dogs that they would likely classify them as anomalies.



Chihuahuas
(Doesn't look like a regular dog)

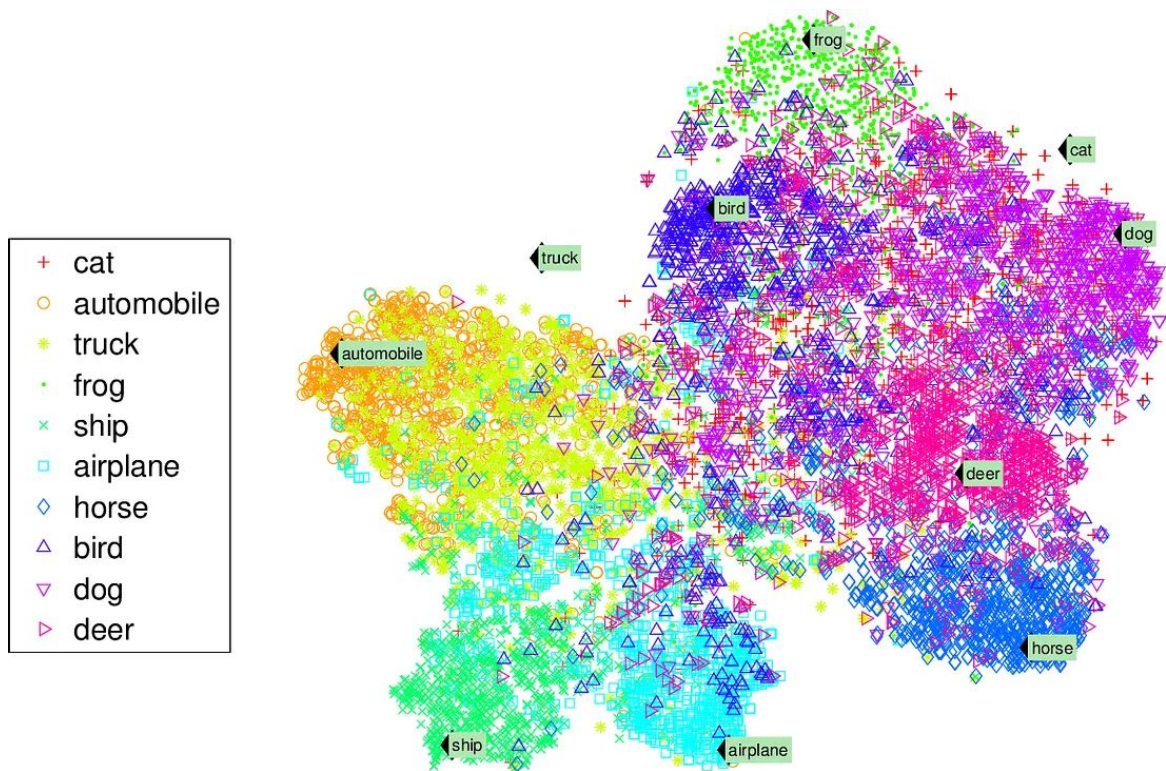
Types of Unsupervised Learning Tasks

2. Anomaly Detection and Novelty Detection:

- One-class SVM
- Isolation Forest
- K-Means Clustering

3. Visualization & Dimensionality Reduction:

- **Visualization algorithms**-You feed them a lot of complex and unlabeled data, and they output a 2D or 3D representation of your data that can easily be plotted.
- These algorithms try to preserve as much structure as they can (e.g., trying to keep separate clusters in the input space from overlapping in the visualization) so that you can understand how the data is organized and perhaps identify unsuspected patterns.



Notice how animals are rather well separated from vehicles and how horses are close to deer but far from birds.

3. Visualization & Dimensionality Reduction:

- **Dimensionality reduction:** - used to **reduce high-dimensional data into lower dimensions** for better visualization or preprocessing.
- Dimensionality reduction reduces the number of input variables in a dataset while retaining its key patterns, helping to simplify models, reduce computation time and improve visualization.
- Goal is to **simplify the data without losing too much information**. One way to do this is to **merge several correlated features into one**.
- For example, a car's mileage may be strongly correlated with its age, so the dimensionality reduction algorithm will merge them into one feature that represents the **car's wear and tear**. This is called **feature extraction**.
- **feature extraction-** You build a new set of features from the original feature set. 29

Types of Unsupervised Learning Tasks

3. Visualization & Dimensionality Reduction:

- Principal Component Analysis (PCA)
- Kernel PCA
- Locally Linear Embedding (LLE)
- t-Distributed Stochastic Neighbor Embedding (t-SNE)

Types of Unsupervised Learning Tasks

4. Association rule learning :

- Goal is to dig into large amounts of data and discover interesting relations between attributes.
- Used to find relationships or associations among a set of items in datasets(often transactional).
- Association rule learning is a machine learning model for discovering interesting relations called association rules between variables in large databases using some measures of “interestingness”.

Types of Unsupervised Learning Tasks

• buy milk. Thus, you may want to place these items close to one another.

- For example, a rule stating

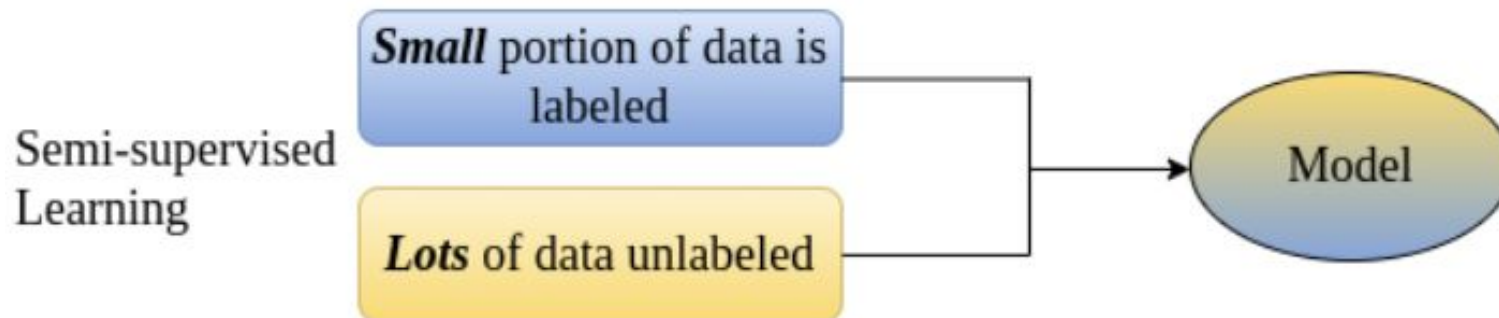
$$\{\mathbf{Bread}, \mathbf{Butter}\} \rightarrow \{\mathbf{Milk}\}$$

- indicates that customers who buy bread and butter also tend to buy milk.

- Apriori algorithm

Semi-supervised Learning

- In semi-supervised learning, the machine learning algorithms are trained on a **small amount of labeled data and a large amount of unlabeled data**.
- Situations where **obtaining a sufficient amount of labeled data is difficult and expensive**, but gathering unlabeled data is much easier.
- In such scenarios, neither fully supervised nor unsupervised learning methods will provide accurate outcomes.
- This is where semi-supervised learning methods can be implemented.

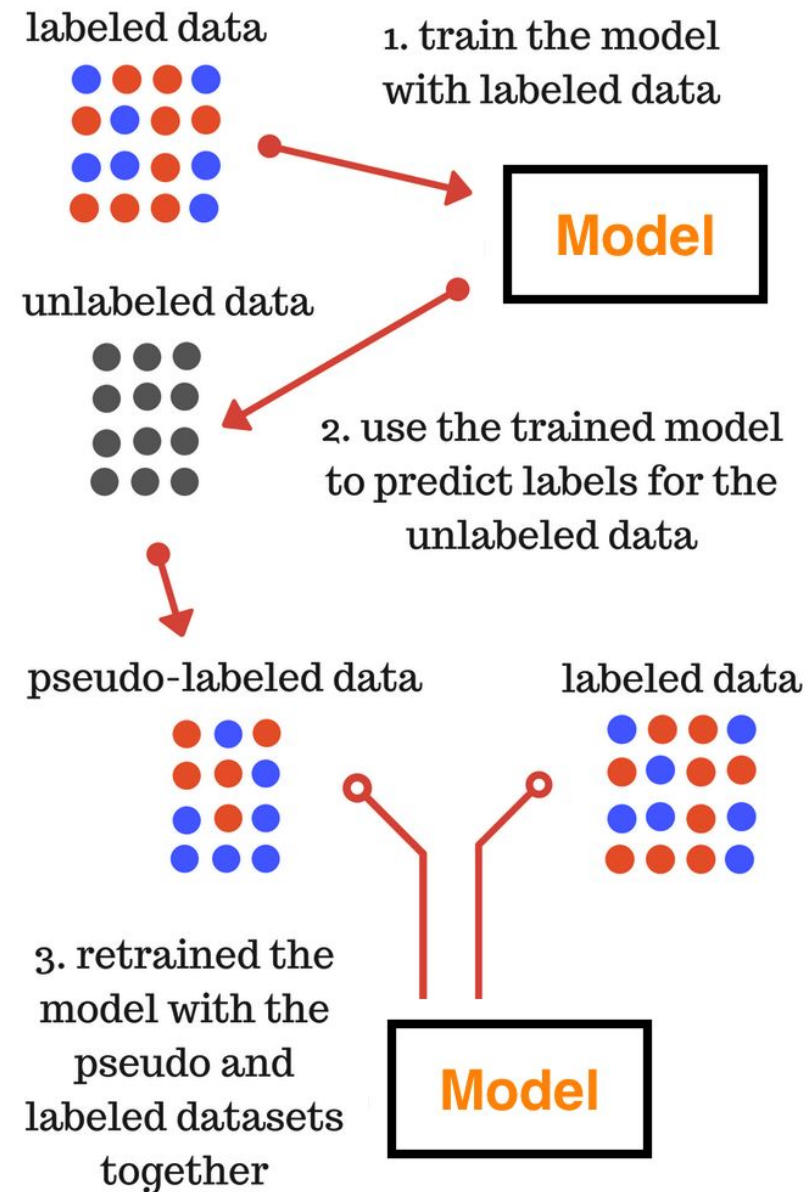


Techniques Used in Semi-Supervised Learning

- There are several techniques for doing SSL, which are described below.

Pseudo-labeling

- Instead of manually labeling the unlabelled data, we give our model approximate labels on the basis of the labelled data.



- This process happens iteratively as the model improves and is able to perform with a greater degree of accuracy.

Semi-supervised Learning

- Most semisupervised learning algorithms are **combinations of unsupervised and supervised algorithms**.
- For example, **deep belief networks (DBNs)** are based on unsupervised components called **restricted Boltzmann machines (RBMs)** stacked on top of one another.
- RBMs are trained sequentially in an unsupervised manner, and then the whole system is fine-tuned using supervised learning techniques.

Reinforcement Learning

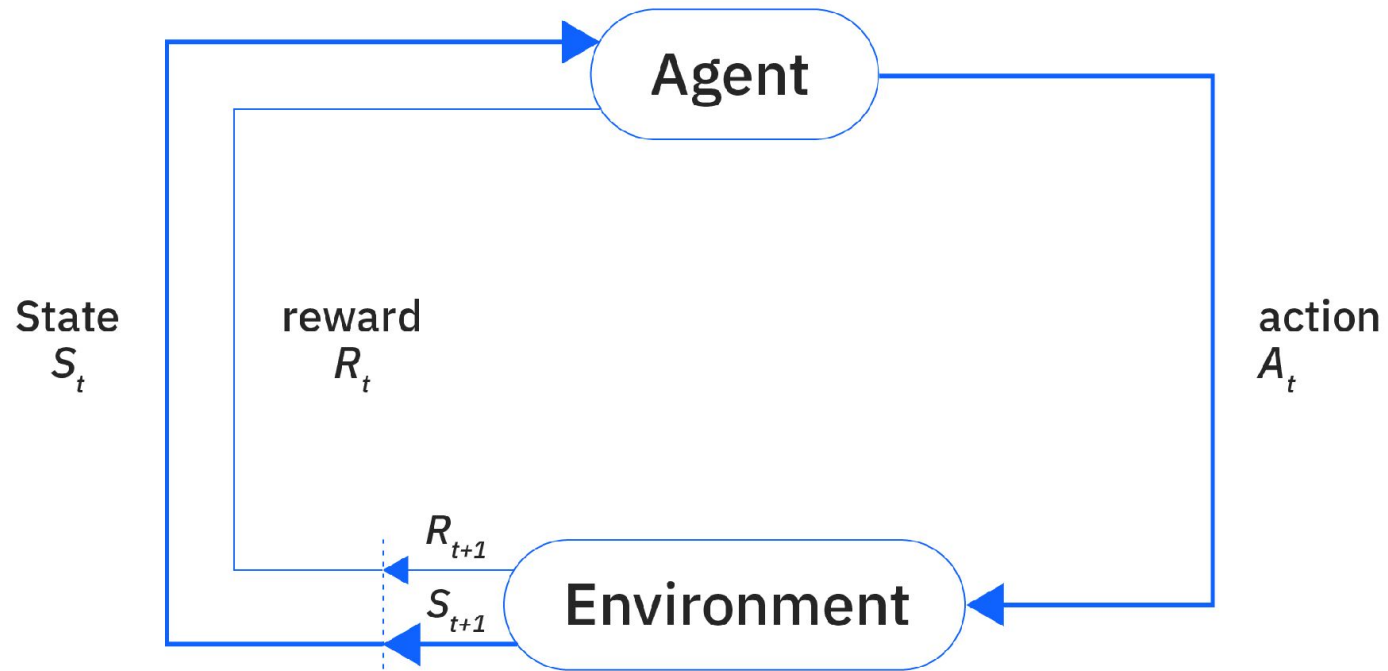
- Reinforcement Learning(RL) is a type of machine learning **where an agent learns to make decisions by interacting with an environment.**
- The agent takes actions, and get rewards in return (or penalties in the form of negative rewards).
- It must then learn by itself what is the best strategy, called a policy, to get the most reward over time.
- A policy defines what action the agent should choose when it is in a given situation.

Reinforcement Learning

The key components are:

1. **Agent** : Learns and makes decision.
2. **Environment** : The world the agent interacts with.
3. **State** : The current situation.
4. **Action** : A move the agent can make.
5. **Reward** : Feedback (good or bad) from the environment.
6. **Policy** : The agent's strategy(what action to take in each state).
7. **Value** : How good a state or action is (in terms of future rewards).

Reinforcement Learning



Reinforcement Learning Model

Reinforcement Learning

- For example, many robots implement Reinforcement Learning algorithms to learn how to walk.
- **DeepMind's AlphaGo program** is also a good example of Reinforcement Learning:
 - it made the headlines in May 2017 when it beat the world champion Ke Jie at the game of Go.
 - It learned its winning policy by analyzing millions of games, and then playing many games against itself.
 - Note that learning was turned off during the games against the champion;
 - AlphaGo was just applying the policy it had learned.

Types of Machine Learning Systems

Supervised Learning	Unsupervised Learning	Semi-Supervised Learning	Reinforcement Learning
Model trained with a labeled data set. For each input, there is a corresponding known output	Model trained with unlabeled data	Model trained with a small labeled data set and a large unlabeled data set	An agent learns to make decisions by interacting with a dynamic environment

Types of machine learning

Parameter Estimation

- Parameter estimation is a statistical method used to estimate unknown parameters from observed data.
- It is widely used in statistics, probability, and machine learning.
- In parameter estimation, unknown model parameters are estimated using sample data.

Definition:

- Parameter estimation is the process of using sample data to estimate the parameters of a probability distribution or model.



Importance:

- Used for prediction and decision making.
- Helps understand data behavior.
- Important in machine learning and artificial intelligence.
- Used in classification and regression models.

Role of Probability Distribution

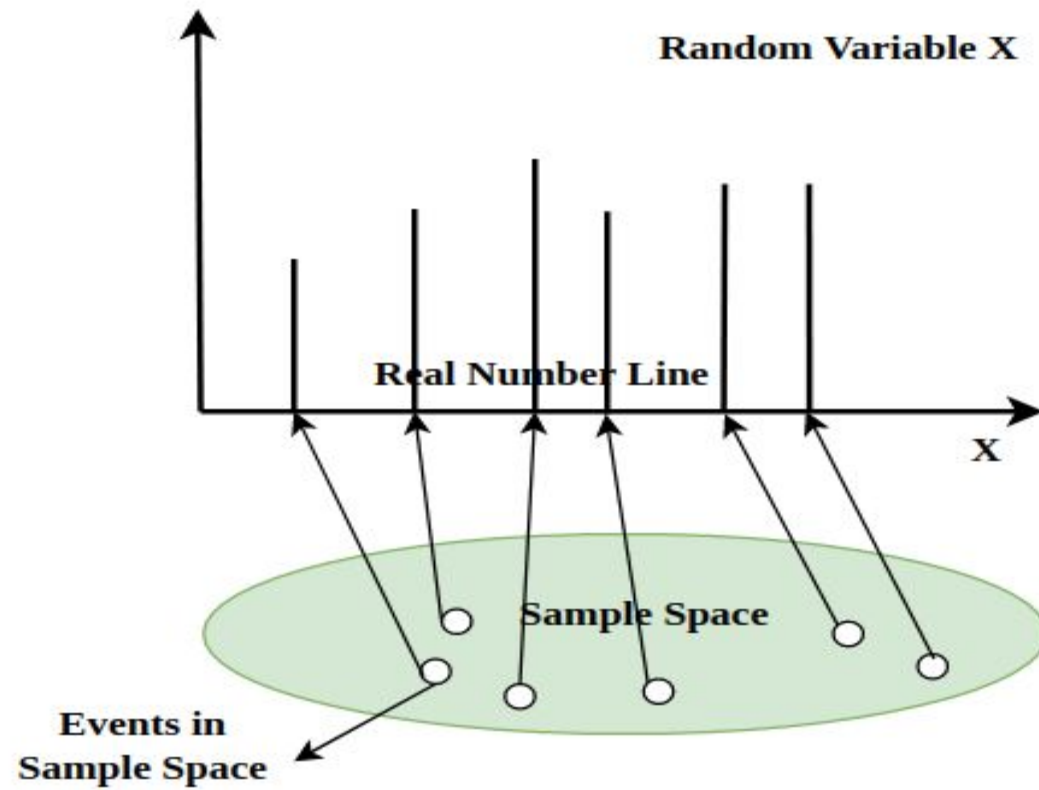
- Probability distribution models random data.
- It describes how likely different outcomes are.
- Parameter estimation depends on probability distributions.

Examples :

- Bernoulli Distribution
- Binomial Distribution
- Poisson Distribution
- Normal Distribution

Probability Distribution

- A probability distribution is a mathematical description of the probabilities of events, subsets of the sample space.
- A probability distribution can be described in various forms, such as by a probability mass function or a cumulative distribution function.



We draw two cards successively with replacement from a well-shuffled deck of 52 cards. Find the **probability distribution** of finding aces.

Let's define a random variable "X", which means number of aces.

Since we are drawing two cards with replacement from a deck of 52 cards, X can only take on the values 0, 1 or 2 as the cards are drawn with replacement, the two draws are independent experiments.

Calculating the probabilities:

$$\begin{aligned} P(X = 0) &= P(\text{both cards are non-aces}) \\ &= P(\text{non-ace}) \times P(\text{non-ace}) \\ &= 48/52 \times 48/52 = 144/169 \end{aligned}$$

$$\begin{aligned} P(X = 1) &= P(\text{one of the cards is ace}) \\ &= P(\text{non-ace and then ace}) + P(\text{ace and then non-ace}) \\ &= P(\text{non-ace}) \times P(\text{ace}) + P(\text{ace}) \times P(\text{non-ace}) \\ &= 48/52 \times 4/52 + 4/52 \times 48/52 = 24/169 \end{aligned}$$

$$\begin{aligned} P(X = 2) &= P(\text{Both the cards are aces}) \\ &= P(\text{ace}) \times P(\text{ace}) \\ &= 4/52 \times 4/52 = 1/169 \end{aligned}$$

Now we have the probability distribution for the discrete random variable X.

It can be represented in the following table:

X	0	1	2
P(X = x)	144/169	24/169	1/169

Discrete Uniform Distribution

- In probability theory and statistics, the **discrete uniform distribution** is a symmetric probability distribution wherein a finite number of values are equally likely to be observed; every one of n values has equal probability $1/n$.
- Another way of saying "discrete uniform distribution" would be "a known, finite number of outcomes equally likely to happen".
- Throwing dice, tossing coin

Bernoulli distribution

- In probability theory and statistics, the Bernoulli distribution, named after Swiss mathematician Jacob Bernoulli, is the discrete probability distribution of a random variable which takes the value 1 with probability p and the value 0 with probability $q=1-p$.
- It can be thought of as a model for the set of possible outcomes of any single experiment that asks a yes–no question.
- Tossing a coin

If X is a random variable with this distribution, then:

$$\Pr(X = 1) = p = 1 - \Pr(X = 0) = 1 - q.$$

The probability mass function f of this distribution, over possible outcomes k , is

$$f(k; p) = \begin{cases} p & \text{if } k = 1, [2] \\ q = 1 - p & \text{if } k = 0. \end{cases}$$

This can also be expressed as

$$f(k; p) = p^k (1 - p)^{1-k} \quad \text{for } k \in \{0, 1\}$$

Applications

- ❑ Some examples of such events are as follows:
 - ❑ a team will win a championship or not,
 - ❑ a student will pass or fail an exam, and
 - ❑ a rolled dice will either show a 6 or any other number.

Example: Can throwing a fair die 50 times be considered an example of 50 Bernoulli trials if we define:

- Success is getting an even number (2, 4, or 6),
- Failure as getting an odd number (1, 3, or 5)

Answer:

Yes, this can be considered as example of 50 Bernoulli trials

- There are 3 even numbers out of 6 possible outcomes, so $p = 3/6 = 1/2$
 - There are 3 odd numbers out of 6, so $q = 3/6 = 1/2$
- So, throwing a fair die 50 times with this definition is a classic example of 50 Bernoulli trials, with $p=1/2$ and $q = 1/2$

To qualify as Bernoulli trials, the following conditions must be met:

- **Only Two Outcomes per Trial:** You've defined "Success" as rolling an even number (2, 4, 6) and "Failure" as rolling an odd number (1, 3, 5). This reduces the outcome space to two categories — satisfying this condition.
- **Constant Probability of Success:** For a fair die:
 - Probability of Success (even number) = $3/6 = 0.5$
 - Probability of Failure (odd number) = $3/6 = 0.5$These probabilities remain constant across all 50 rolls.
- **Independence of Trials:** Each die roll is independent of the others — the outcome of one roll does not affect the next.

So, throwing a fair die 50 times with this definition is a classic example of 50 Bernoulli trials, with $p=1/2$ and $q = 1/2$

Binomial distribution

- ❑ In probability theory and statistics, the binomial distribution with parameters n and p is the discrete probability distribution of the number of successes in a sequence of n independent experiments, each asking a yes–no question, and each with its own Boolean-valued outcome:
 - ❑ *success* (with probability p) or
 - ❑ *failure* (with probability $q = 1 - p$).
- ❑ A single success/failure experiment is also called a Bernoulli trial or Bernoulli experiment, and a sequence of outcomes is called a Bernoulli process; for a single trial, i.e., $n = 1$, the binomial distribution is a Bernoulli distribution.

Binomial distribution

The **binomial distribution** models the number of successes (x) in n independent Bernoulli trials, each with success probability p .

For example,

For **1 success in 6 trials**, there are 6 possible sequences (e.g., PQQQQQ, QPQQQQ,.....), each with probability $p \cdot (1-p)^5$

Therefore the total Probability is given as $= 6 \cdot p \cdot (1-p)^5$

Generalizing the idea, if Y is a Binomial Random Variable, the Probability Function $P(Y)$ for the Binomial Distribution for n number of trials is given as:

$$P(Y) = {}^nC_x p^x (1-p)^{n-x}$$

where

- p is the probability of success in a given trial,
- ' x ' be the number of successes, $x = 0, 1, 2, \dots, n$

● Probability mass function

In general, if the random variable X follows the binomial distribution with parameters $n \in \mathbb{N}$ and $p \in [0, 1]$, we write $X \sim B(n, p)$. The probability of getting exactly k successes in n independent Bernoulli trials is given by the probability mass function:

$$f(k, n, p) = \Pr(k; n, p) = \Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

for $k = 0, 1, 2, \dots, n$, where

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

● is the binomial coefficient.

- The formula can be understood as follows: k successes occur with probability p^k and $n - k$ failures occur with probability $(1 - p)^{n - k}$.
- However, the k successes can occur anywhere among the n trials, and there are $\text{binom}\{\{n\}\{k\}\}$ different ways of distributing k successes in a sequence of n trials.

Example: When a fair coin is tossed 10 times, find the probability of getting i. exactly six heads. ii. at least six heads.

Answer: Probability of exactly six heads

Given:

$$P(X = 6) = \binom{10}{6} \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^4$$
$$P(X = 6) = \binom{10}{6} \left(\frac{1}{2}\right)^{10}$$
$$\binom{10}{6} = \frac{10!}{6!4!} = 210$$
$$\therefore P(X = 6) = 210 \times \left(\frac{1}{1024}\right) = \frac{210}{1024} = 0.2051$$

(ii) Probability of at least six heads

“At least six heads” means $X = 6, 7, 8, 9, 10$.

$$P(X \geq 6) = \sum_{r=6}^{10} \binom{10}{r} \left(\frac{1}{2}\right)^{10}$$
$$P(X \geq 6) = \frac{1}{1024} \left[\binom{10}{6} + \binom{10}{7} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10} \right]$$
$$= \frac{1}{1024} [210 + 120 + 45 + 10 + 1] = \frac{386}{1024} = 0.3770$$

$$\boxed{P(X \geq 6) = 0.3770}$$

Applications

- find the probability that more than a certain number of patients in a random sample of 100 will experience negative side effects.
- to model the probability that a certain number of credit card transactions are fraudulent.
- to model the probability that a certain number of spam emails land in an inbox per day.
- to model the probability that rivers overflow a certain number of times each year due to excessive rain.

Poisson distribution

- In probability theory and statistics, the **Poisson distribution** is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space if these events occur with a known constant mean rate and independently of the time since the last event.

Probability mass function

A discrete random variable X is said to have a Poisson distribution, with parameter $\lambda > 0$ if it has a probability mass function given by

$$f(k; \lambda) = \Pr(X=k) = \frac{\lambda^k e^{-\lambda}}{k!},$$

where λ (lambda) represents the *average rate* of occurrence in a given interval

- k is the number of occurrences ($k = 0, 1, 2 \dots$)
- e is Euler's number ($e = 2.71828\dots$)
- $!$ is the factorial function.

The positive real number λ is equal to the expected value of X and also to its variance.

$$\lambda = E(X) = \text{Var}(X).$$

Example: A bakery sells an average of 5 cupcakes per hour. What's the probability the sell exactly 3 cupcakes in the next hour?

$\lambda=5$ (average rate), $k=3$ (desired events).

$$P(X = x) = \frac{e^{-\lambda} \lambda^k}{x!}$$
$$P(X = 3) = \frac{e^{-5} 5^3}{3!} \approx 0.14$$

Applications

- ❑ The random variables that follow a Poisson distribution are as follows:
- ❑ To count the number of defects of a finished product
- ❑ To count the number of deaths in a country by any disease or natural calamity
- ❑ To count the number of infected plants in the field
- ❑ To count the number of bacteria in the organisms or the radioactive decay in atoms
- ❑ To calculate the waiting time between the events.
- ❑ To model the number of expected calls per hour in a call centre
- ❑ To model the number of expected customers that will arrive at the restaurant per day.

Exponential Distribution

Probability density function

The probability density function (pdf) of an exponential distribution is

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

Here $\lambda > 0$ is the parameter of the distribution, often called the *rate parameter*. The distribution is supported on the interval $[0, \infty)$. If a random variable X has this distribution, we write $X \sim \text{Exp}(\lambda)$.

Normal Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$f(x)$ = probability density function

σ = standard deviation

μ = mean

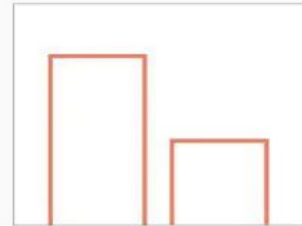
Types of probability distribution in machine learning



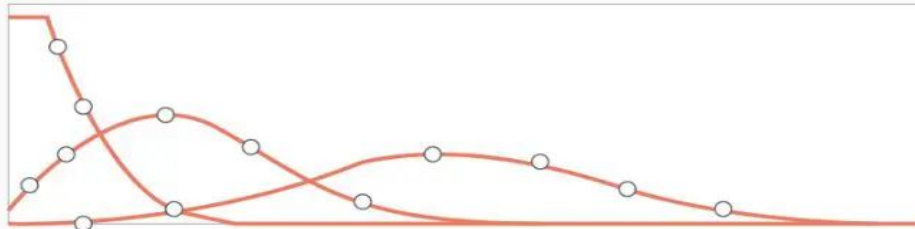
Uniform distribution



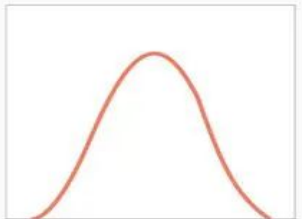
Binomial distribution



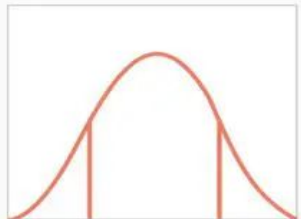
Bernoulli distribution



Poisson distribution



Normal distribution



T-test distribution



Exponential distribution

Question 1: A box contains 4 blue balls and 3 green balls. Find the probability distribution of the number of green balls in a random draw of 3 balls.

Solution:

Given that the total number of balls is 7 out of which 3 have to be drawn at random. On drawing 3 balls the possibilities are all 3 are green, only 2 is green, only 1 is green, and no green. Hence $X = 0, 1, 2, 3$.

- $P(\text{No ball is green}) = P(X = 0) = {}^4C_3 / {}^7C_3 = 4/35$
- $P(1 \text{ ball is green}) = P(X = 1) = {}^3C_1 \times {}^4C_2 / {}^7C_3 = 18/35$
- $P(2 \text{ balls are green}) = P(X = 2) = {}^3C_2 \times {}^4C_1 / {}^7C_3 = 12/35$
- $P(\text{All 3 balls are green}) = P(X = 3) = {}^3C_3 / {}^7C_3 = 1/35$

Hence, the probability distribution for this problem is given as follows

X	0	1	2	3
$P(X)$	4/35	18/35	12/35	1/35

Questions

- A coin is tossed 12 times. What is the probability of getting exactly 7 heads?

● Solution:

- Given that a coin is tossed 12 times. (i.e) $n = 12$
- Thus, a probability pf getting head in single toss = $\frac{1}{2}$.
- (i.e) $p = \frac{1}{2}$. So, $1-p = 1-\frac{1}{2} = \frac{1}{2}$.
- We know that the binomial probability distribution is
- $P(r) = {}^nC_r \cdot p^r (1-p)^{n-r}$.
- Now, we have to find the probability of getting exactly 7 heads.(i.e) $r = 7$.
- Substituting the values in the binomial distribution formula, we get
- $P(7) = {}^{12}C_7 \cdot (\frac{1}{2})^7 (\frac{1}{2})^{12-7}$
- $P(7) = 792 \cdot (\frac{1}{2})^7 (\frac{1}{2})^5$
- $P(7) = 792 \cdot (\frac{1}{2})^{12}$
- $P(7) = 792 (1/4096)$
- $P(7) = 0.193$
- Therefore, the probability of getting exactly 7 heads is 0.193.

- The probability that a person can achieve a target is $\frac{3}{4}$. The count of tries is 5. What is the probability that he will attain the target at least thrice?

● Solution:

- Given that, $p = \frac{3}{4}$, $q = \frac{1}{4}$, $n = 5$.
- Using binomial distribution formula, we get
- $P(X) = {}^nC_x \cdot p^x (1 - p)^{n-x}$
- Thus, the required probability is:
- $P(X = 3) + P(X=4) + P(X=5)$
- $= {}^5C_3 \cdot \left(\frac{3}{4}\right)^3 \left(\frac{1}{4}\right)^2 + {}^5C_4 \cdot \left(\frac{3}{4}\right)^4 \left(\frac{1}{4}\right)^1 + {}^5C_5 \cdot \left(\frac{3}{4}\right)^5$
- $= 459/512.$
- Therefore, the probability that the person will attain the target at least thrice is $459/512.$

Types of Parameter Estimation

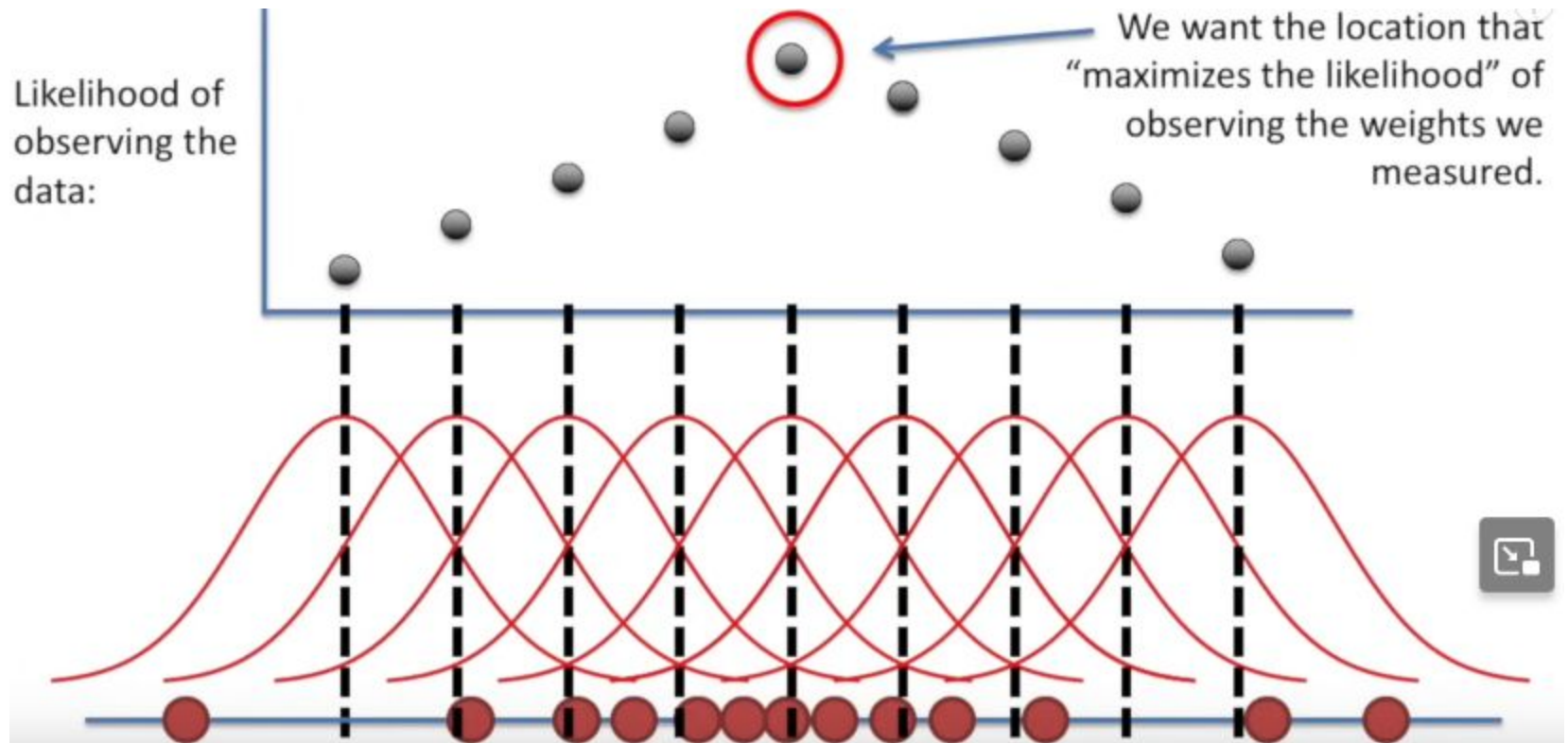
- 1. Maximum Likelihood Estimation (MLE)
- 2. Maximum A Posteriori Estimation (MAP)

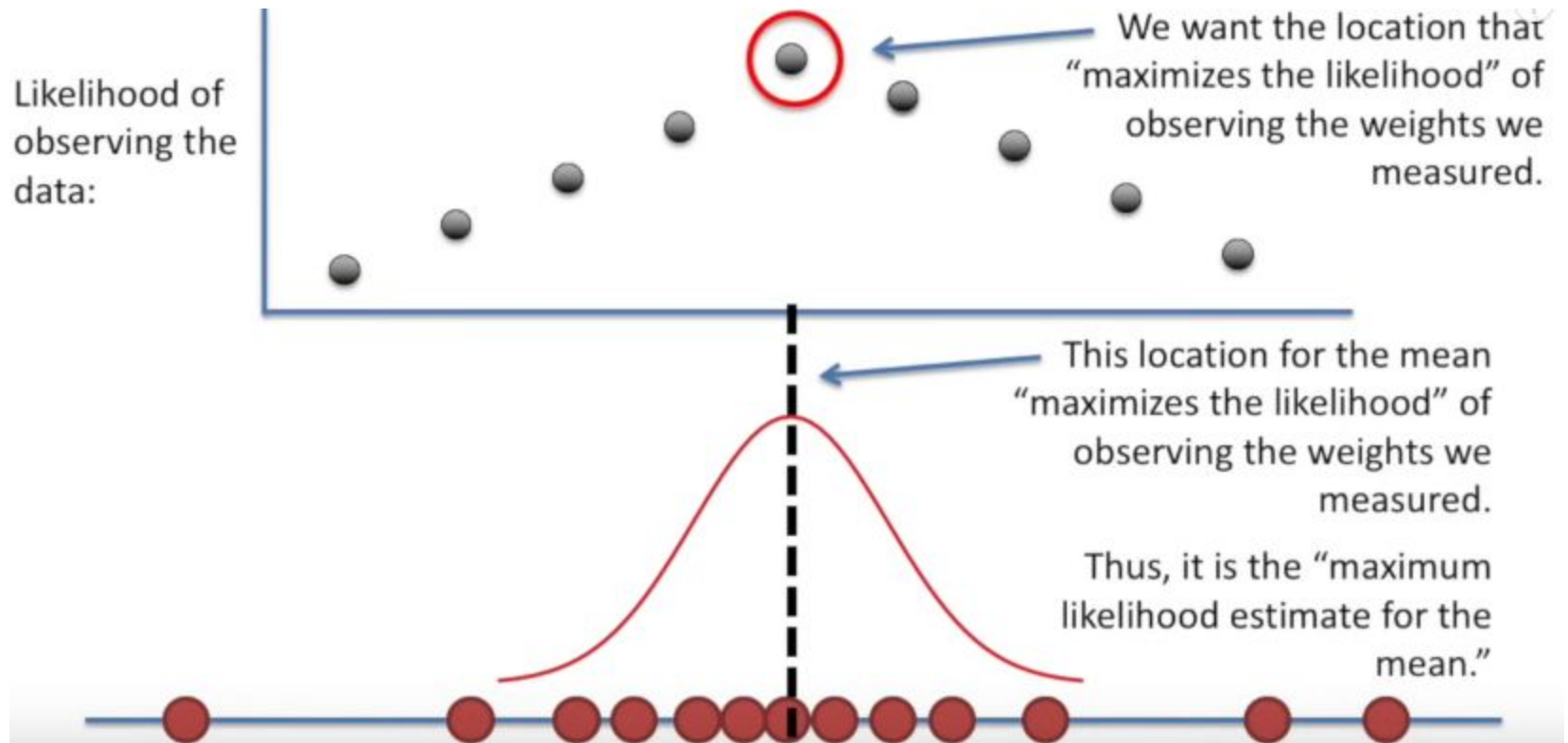
Likelihood function

- The probability of data given the parameter is commonly referred to as the likelihood.

Goal of maximum likelihood is to find an optimal way to fit a distribution to a data

This process, called **Maximum Likelihood Estimation** (MLE), finds the best-fitting distribution to data by selecting parameter values that make the observed data most likely to occur.





Maximum likelihood estimation

- **Maximum likelihood estimation (MLE)** is a method of estimating the parameters of an assumed probability distribution, given some observed data.
- This is achieved by maximizing a likelihood function so that, under the assumed statistical model, the observed data is most probable.
- The point in the parameter space that maximizes the likelihood function is called the maximum likelihood estimate.

- ML estimation tries to find the estimate of the parameter θ by maximizing the likelihood function.
- Assume we have independent and identically distributed (i.i.d) random samples $x_1, x_2, \dots, x_n$ that follow a distribution $f(x_1, x_2, \dots, x_n; \theta)$, which depends on the unknown parameter θ .
- The goal is to estimate this unknown quantity such that it maximizes the probability of observing this random sample.

Definition 1. *The likelihood function is the density function regarded as a function of θ .*

$$\mathbf{L}(\theta|\mathbf{x}) = \mathbf{f}(\mathbf{x}|\theta), \theta \in \Theta.$$

The maximum likelihood estimator (MLE),

$$\hat{\theta}(\mathbf{x}) = \arg \max_{\theta} \mathbf{L}(\theta|\mathbf{x}).$$

1. Construct the likelihood function $L(\theta|x) = f(x_1, x_2, \dots, x_n; \theta)$
2. Use the property of i.i.d random samples to break the joint PDF to the product of n marginal PDF.

$$f(x_1, x_2, \dots, x_n; \theta) = \prod_{i=1}^n f(x_i; \theta)$$

3. Take a logarithm (usually a natural log) to change the product to summation. This does not change the optimal estimator since the logarithm is a monotonic function.

$$LL(\theta; x) = \sum_{i=1}^n \log f(x_i; \theta)$$

Where $LL(\theta; x)$ represents the log-likelihood function.

4. Differentiate the log-likelihood function with respect to θ and set it to zero.

$$\frac{\partial LL(\theta; x)}{\partial \theta} = 0$$

5. The estimator will only be a function of observed data.

$$\hat{\theta}_{ML} = g(x_1, x_2, \dots, x_n)$$

- A coin is tossed 100 times and lands heads 55 times. What is the maximum likelihood estimate for p , *the probability of heads*?
 - **Experiment**: Flip the coin 100 times and count the number of heads.
 - **Data**: The data is the result of the experiment. In this case it is '55 heads'.
 - **Parameter(s) of interest**: We are interested in the value of the unknown parameter p .
 - **Likelihood**, or **likelihood function**: this is $P(\text{data} | p)$. Note it is a function of both the data and the parameter p . In this case the likelihood is

$$P(55 \text{ heads} | p) = \binom{100}{55} p^{55} (1 - p)^{45}.$$

The likelihood $P(\text{data} | p)$ changes as the parameter of interest p changes.

- The maximum likelihood estimate (MLE) for the parameter p is the value of p that maximizes the likelihood $P(\text{data} \mid p)$. That is, the MLE is the value of p for which the data is most likely.

answer: We had the likelihood $P(55 \text{ heads} \mid p) = \binom{100}{55} p^{55} (1 - p)^{45}$. Therefore the log likelihood is

$$\ln(P(55 \text{ heads} \mid p)) = \ln \left(\binom{100}{55} \right) + 55 \ln(p) + 45 \ln(1 - p).$$

Maximizing likelihood is the same as maximizing log likelihood. We check that calculus gives us the same answer as before:

$$\begin{aligned} \frac{d}{dp}(\log \text{ likelihood}) &= \frac{d}{dp} \left[\ln \left(\binom{100}{55} \right) + 55 \ln(p) + 45 \ln(1 - p) \right] \\ &= \frac{55}{p} - \frac{45}{1 - p} = 0 \\ &\Rightarrow 55(1 - p) = 45p \\ &\Rightarrow \hat{p} = .55 \end{aligned}$$

Light bulbs

Suppose that the lifetime of *Badger* brand light bulbs is modeled by an exponential distribution with (unknown) parameter λ . We test 5 bulbs and find they have lifetimes of 2, 3, 1, 3, and 4 years, respectively. What is the MLE for λ ?

answer: We need to be careful with our notation. With five different values it is best to use subscripts. Let X_i be the lifetime of the i^{th} bulb and let x_i be the value X_i takes. Then each X_i has pdf $f_{X_i}(x_i) = \lambda e^{-\lambda x_i}$. We assume the lifetimes of the bulbs are independent, so the joint pdf is the product of the individual densities:

$$f(x_1, x_2, x_3, x_4, x_5 | \lambda) = (\lambda e^{-\lambda x_1})(\lambda e^{-\lambda x_2})(\lambda e^{-\lambda x_3})(\lambda e^{-\lambda x_4})(\lambda e^{-\lambda x_5}) = \lambda^5 e^{-\lambda(x_1+x_2+x_3+x_4+x_5)}.$$

Note that we write this as a conditional density, since it depends on λ . Viewing the data as fixed and λ as variable, this density is the likelihood function. Our data had values

$$x_1 = 2, x_2 = 3, x_3 = 1, x_4 = 3, x_5 = 4.$$

So the likelihood and log likelihood functions with this data are

$$f(2, 3, 1, 3, 4 | \lambda) = \lambda^5 e^{-13\lambda}, \quad \ln(f(2, 3, 1, 3, 4 | \lambda)) = 5 \ln(\lambda) - 13\lambda$$

Finally we use calculus to find the MLE:

$$\frac{d}{d\lambda}(\log \text{ likelihood}) = \frac{5}{\lambda} - 13 = 0 \Rightarrow \boxed{\hat{\lambda} = \frac{5}{13}}.$$

Example : Suppose that X is a discrete random variable with the following probability mass function: where $0 \leq \theta \leq 1$ is a parameter. The following 10 independent observations

X	0	1	2	3
$P(X)$	$2\theta/3$	$\theta/3$	$2(1-\theta)/3$	$(1-\theta)/3$

were taken from such a distribution: (3,0,2,1,3,2,1,0,2,1). What is the maximum likelihood estimate of θ .

Solution: Since the sample is (3,0,2,1,3,2,1,0,2,1), the likelihood is

$$\begin{aligned} L(\theta) = & P(X = 3)P(X = 0)P(X = 2)P(X = 1)P(X = 3) \\ & \times P(X = 2)P(X = 1)P(X = 0)P(X = 2)P(X = 1) \end{aligned}$$

Substituting from the probability distribution given above, we have

$$L(\theta) = \prod_{i=1}^n P(X_i|\theta) = \left(\frac{2\theta}{3}\right)^2 \left(\frac{\theta}{3}\right)^3 \left(\frac{2(1-\theta)}{3}\right)^3 \left(\frac{1-\theta}{3}\right)^2$$

Let us look at the log likelihood function

$$\begin{aligned}l(\theta) &= \log L(\theta) = \sum_{i=1}^n \log P(X_i|\theta) \\&= 2 \left(\log \frac{2}{3} + \log \theta \right) + 3 \left(\log \frac{1}{3} + \log \theta \right) + 3 \left(\log \frac{2}{3} + \log(1 - \theta) \right) + 2 \left(\log \frac{1}{3} + \log(1 - \theta) \right) \\&= C + 5 \log \theta + 5 \log(1 - \theta)\end{aligned}$$

where C is a constant which does not depend on θ . It can be seen that the log likelihood function is easier to maximize compared to the likelihood function.

Let the derivative of $l(\theta)$ with respect to θ be zero:

$$\frac{dl(\theta)}{d\theta} = \frac{5}{\theta} - \frac{5}{1 - \theta} = 0$$

and the solution gives us the MLE, which is $\hat{\theta} = 0.5$.

Example : The Pareto distribution has been used in economics as a model for a density function with a slowly decaying tail:

$$f(x|x_0, \theta) = \theta x_0^\theta x^{-\theta-1}, \quad x \geq x_0, \quad \theta > 1$$

Assume that $x_0 > 0$ is given and that $X_1, X_2, \dots, X_n$ is an i.i.d. sample. Find the MLE of θ .

Solution: The log-likelihood function is

$$\begin{aligned}l(\theta) &= \sum_{i=1}^n \log f(X_i|\theta) = \sum_{i=1}^n (\log \theta + \theta \log x_0 - (\theta + 1) \log X_i) \\&= n \log \theta + n\theta \log x_0 - (\theta + 1) \sum_{i=1}^n \log X_i\end{aligned}$$

Let the derivative with respect to θ be zero:

$$\frac{dl(\theta)}{d\theta} = \frac{n}{\theta} + n \log x_0 - \sum_{i=1}^n \log X_i = 0$$

Solving the equation yields the MLE of θ :

$$\hat{\theta}_{MLE} = \frac{1}{\overline{\log X} - \log x_0}$$

Exercise 1: Let $X_1, \dots, X_n$ be an i.i.d. sample from a Poisson distribution with parameter λ , i.e.,

$$P(X = x|\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}.$$

Please find the MLE of the parameter λ .

Exercise 2: Let $X_1, \dots, X_n$ be an i.i.d. sample from an exponential distribution with the density function

$$f(x|\beta) = \frac{1}{\beta} e^{-\frac{x}{\beta}}, \text{ with } 0 \leq x < \infty.$$

Please find the MLE of the parameter β .

Exercise 3: Gamma distribution has a density function as

$$f(x|\alpha, \lambda) = \frac{1}{\Gamma(\alpha)} \lambda^\alpha x^{\alpha-1} e^{-\lambda x}, \text{ with } 0 \leq x < \infty.$$

Suppose the parameter α is known, please find the MLE of λ based on an i.i.d. sample $X_1, \dots, X_n$.

Exercise 4: Suppose that $X_1, \dots, X_n$ form a random sample from a distribution for which the pdf $f(x|\theta)$ is as follows:

$$f(x|\theta) = \begin{cases} \theta x^{\theta-1}, & \text{for } 0 < x < 1 \\ 0, & \text{for } x \leq 0 \end{cases}$$

Also suppose that the value of θ is unknown ($\theta > 0$). Find the MLE of θ .

Exercise 5: Suppose that $X_1, \dots, X_n$ form a random sample from a distribution for which the pdf $f(x|\theta)$ is as follows:

$$f(x|\theta) = \frac{1}{2} e^{-|x-\theta|} \quad \text{for } -\infty < x < \infty$$

Also suppose that the value of θ is unknown ($-\infty < \theta < \infty$). Find the MLE of θ .

Maximum a Posteriori (MAP) Estimation

There is another way to estimate a particular parameter value *if you have prior knowledge about the distribution*.

- **Maximum-a-Posteriori Estimation (MAP):** Choose θ that maximizes the **posterior probability** of θ (i.e., probability **in the light of the observed data**)
- **Posterior probability** of θ is given by the Bayes Rule

$$P(\theta | \mathcal{D}) = \frac{P(\theta)P(\mathcal{D} | \theta)}{P(\mathcal{D})}$$

- $P(\theta)$: **Prior probability** of θ (without  having seen any data)
- $P(\mathcal{D} | \theta)$: **Likelihood**
- $P(\mathcal{D})$: Probability of the data (independent of θ)

$$P(\mathcal{D}) = \int P(\theta)P(\mathcal{D} | \theta)d\theta \quad (\text{sum over all } \theta\text{'s})$$

- The Bayes Rule lets us **update our belief** about θ in the light of observed data
- While doing MAP, we usually maximize the **log of the posterior probability**

Maximum a Posteriori (MAP) Estimation

- MAP estimation tries to find the estimate of the parameter θ by maximizing the posterior distribution.
- Maximize the product of likelihood and the prior probability.

$$P(\theta|x) = \frac{P(x|\theta)P(\theta)}{P(x)}$$

- **MAP:** *"tries to find the best setting given the parameters"*
- **MLE:** *"tries to maximize the parameters to best explain the data"*

MAP

Steps:

- Construct the posterior distribution.
- Use the property of random sample to simplify the posterior distribution.
- $$P(\theta|x_1, x_2, \dots, x_n) = P(\theta) \prod_{i=1}^{i=n} P(x_i|\theta)$$
- Take the logarithm (Usually a natural log) to further simplify the above relationship.

$$\log P(\theta|x_1, x_2, \dots, x_n) \propto \log P(\theta) + \sum_{i=1}^{i=n} \log P(x_i|\theta)$$

- Differentiate the above equation with respect to θ and set it to zero

$$\frac{\partial \log P(\theta|x_1, x_2, \dots, x_n)}{\partial \theta} = \frac{\partial \log P(\theta)}{\partial \theta} + \frac{\partial \sum_{i=1}^n \log P(x_i|\theta)}{\partial \theta} = 0$$

- Solve for theta.

- Same as MLE except the **extra log-prior-distribution term!**
- MAP allows incorporating our **prior knowledge** about θ in its estimation

Key Difference Between MLE & MAP

Feature	MLE	MAP
Full Form	Maximum Likelihood Estimation	Maximum A Posteriori
Uses Prior Knowledge	No	Yes
Based On	Likelihood	Posterior Probability
Bayesian Method	No	Yes
Overfitting	More possible	Reduced
Accuracy with Small Data	Less reliable	Better
Data Requirement	Large dataset preferred	Works well with small dataset
Complexity	Simpler	Slightly complex

Bayesian Formulation

Assume we have two random variables A and B.

- A principle rule of probability theory known as the chain rule allows us to specify the joint probability of A and B taking on particular values a and b, $P(a, b)$, as the product of the conditional probability that A will take on value 'a' given that B has taken on value b, $P(a|b)$, and the marginal probability that B takes on value b, $P(b)$.
- That is, Joint probability = Conditional Probability
x Marginal Probability
- Marginal probability is the probability of a single event occurring, without considering any other events.
- $P(a, b) = P(a|b)P(b)$
- Also, $P(a, b) = P(b|a)P(a)$

- Combining both,
- $P(a|b)P(b) = P(b|a)P(a)$

$$P(a|b) = \frac{P(b|a)P(a)}{P(b)}$$

Bayes' Theorem

posterior

likelihood prior

$$P(H|\mathbf{X}) = \frac{P(\mathbf{X}|H)P(H)}{P(\mathbf{X})}$$

evidence

$\mathbf{X}$ - data sample ("*evidence*").

H - *hypothesis* that $\mathbf{X}$ belongs to class C .

$P(\mathbf{X})$: probability that sample data is observed.

$P(H)$ - *prior probability*: the initial probability.

$P(H|\mathbf{X})$ - *posteriori probability*: the probability that the hypothesis holds given the observed data sample $\mathbf{X}$.

$P(\mathbf{X}|H)$ (likelihood): the probability of observing the sample $\mathbf{X}$, given that the hypothesis holds.

Bayesian Classification: Why?

- A statistical classifier: performs *probabilistic prediction*, i.e., predicts class membership probabilities.
- Foundation: Based on Bayes' Theorem.
- Incremental: Each training example can incrementally increase/decrease the probability that a hypothesis is correct — prior knowledge can be combined with observed data.

Classification - Derive the Maximum Posteriori

- Let D be a training set of tuples and their associated class labels, and each tuple is represented by an n -D attribute vector $\mathbf{X} = (x_1, x_2, \dots, x_n)$.
- Suppose there are m classes $C_1, C_2, \dots, C_m$.
- Classification is to derive the maximum posteriori, i.e., the maximal $P(C_i|\mathbf{X})$. This can be derived from Bayes' theorem,

$$P(C_i|\mathbf{X}) = \frac{P(\mathbf{X}|C_i)P(C_i)}{P(\mathbf{X})}$$

$$P(C_i) \geq 0 \text{ and } \sum_{i=1}^K P(C_i) = 1$$

choose C_i if $P(C_i | \mathbf{x}) = \max_k P(C_k | \mathbf{x})$

- A simplified assumption: attributes are conditionally independent (i.e., no dependence relation between attributes):

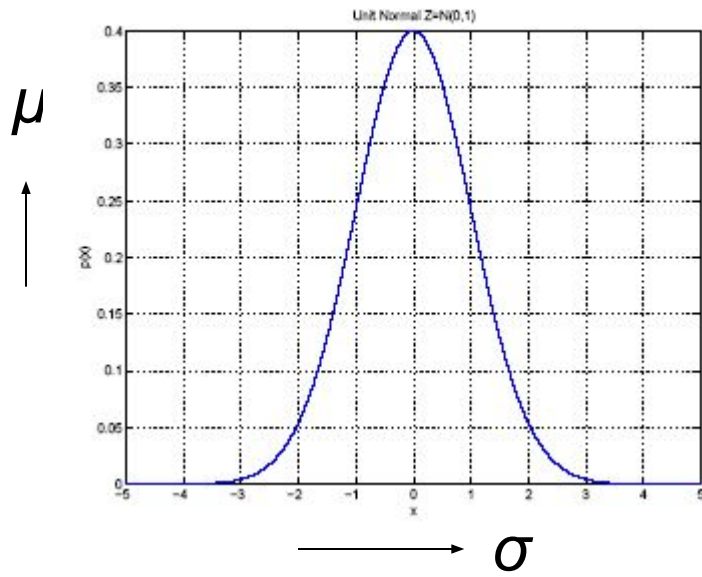
$$\begin{aligned} P(\mathbf{X} | C_i) &= \prod_{k=1}^n P(x_k | C_i) \\ &= P(x_1 | C_i) \times P(x_2 | C_i) \times \dots \times P(x_n | C_i) \end{aligned}$$

Bayes' Rule: $K > 2$ Classes

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$$\begin{aligned} P(C_i | \mathbf{x}) &= \frac{p(\mathbf{x} | C_i)P(C_i)}{p(\mathbf{x})} \\ &= \frac{p(\mathbf{x} | C_i)P(C_i)}{\sum_{k=1}^K p(\mathbf{x} | C_k)P(C_k)} \end{aligned}$$

Gaussian (Normal) Distribution



- $p(x) = N(\mu, \sigma^2)$

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

- MLE for μ and σ^2 :

$$m = \frac{\sum x^t}{N}$$

$$s^2 = \frac{\sum (x^t - m)^2}{N}$$

Feature Representation

Definition

- Feature representation is the process of converting raw data into numerical form so that machine learning algorithms can understand and process it.
- A feature is an individual measurable property of data.

Example of Feature Representation

Application	Features
House Price Prediction	Area, Bedrooms, Location
Student Performance Prediction	Attendance, Marks, Assignments
Email Spam Detection	Spam words, Links, Attachments
Image Recognition	Pixels, Shapes, Edges

Feature Extraction

Feature extraction is the process of extracting important information from raw data. Examples:

- Edge detection in images
- Keyword extraction from text
- Frequency extraction from signals

Feature Selection

Feature selection is the process of selecting important features and removing irrelevant features.

Importance of Feature Representation

Advantage	Explanation
Better Accuracy	Helps model learn effectively
Faster Training	Reduces computation
Reduces Overfitting	Removes unnecessary features
Better Generalization	Performs well on unseen data

Problem Formulation in Supervised Learning

Problem formulation is the process of defining:

- Input features
- Output labels
- Learning objective
- Evaluation method

Components of Problem Formulation

Component	Description
Input Space	Set of input features
Output Space	Predicted output
Dataset	Example of Loss Calculation
Learning Function	
Evaluation Metric	

Actual Output (y)	Predicted Output ($\hat{y}$)	Error
100	90	10
50	45	5
80	70	10

Loss function calculates the overall prediction error from all samples.

Loss Function Definition

- A loss function measures the difference between: Actual output

Predicted output

- It indicates how well or poorly the model performs.
- Lower loss value means better prediction accuracy.

Mathematical Representation

$$L(y, \hat{y})$$

$$L(y, \hat{y})$$

Where:

- $y \rightarrow$ Actual output
- $\hat{y} \rightarrow$ Predicted output

Table Representation of Loss Functions for Regression

Loss Function	Definition	Error Calculation Method	Sensitivity to Outliers	Advantages	Disadvantages	Applications	Characteristics	🔍
Mean Squared Error (MSE)	Measures average squared difference between actual and predicted values	Squares each error before averaging	High	Penalizes large errors strongly, smooth for optimization	Sensitive to outliers	House price prediction, forecasting	Produces larger penalty for large errors	
Mean Absolute Error (MAE)	Measures average absolute difference between actual and predicted values	Uses absolute values of errors	Low	Easy to understand, robust to outliers	Not differentiable at zero	Temperature prediction, sales prediction	Treats all errors equally	
Root Mean Squared Error (RMSE)	Square root of MSE	Takes square root after averaging squared errors	High	Error value remains in original unit	Sensitive to large errors	Model evaluation	Indicates prediction accuracy clearly	
Huber Loss	Combination of MSE and MAE	Uses MSE for small errors and MAE for large errors	Moderate	Robust and stable	More computational complexity	Robust regression	Balances MSE and MAE	
Log-Cosh Loss	Smooth approximation of MAE	Uses logarithm of hyperbolic cosine	Low to Moderate	Smooth and differentiable	Computationally expensive	Deep learning regression	Handles errors smoothly	

Loss Functions for Classification

Loss Function	Definition	Error Calculation Method	Advantages	Disadvantages	Applications	Characteristics
Binary Cross Entropy Loss	Measures error in binary classification problems	Compares actual and predicted probabilities	Effective for binary classification	Sensitive to incorrect confident predictions	Spam detection, Disease prediction	Used when output has two classes
Categorical Cross Entropy Loss	Measures error in multi-class classification problems	Calculates probability difference across multiple classes	Suitable for multi-class problems	Computationally expensive for many classes	Image classification, Digit recognition	Used with softmax output

Optimization

Optimization is the process of finding the best model parameters that minimize the loss function.

Roles of Optimization

Role	Explanation
Reduces Error	Minimizes loss value
Improves Predictions	Adjusts parameters for better accuracy
Speeds Learning	Helps model converge faster
Finds Best Parameters	Identifies optimal weights
Enhances Performance	Produces efficient model

Gradient Descent

- Gradient Descent is the most common optimization algorithm used in machine learning.
- It updates model parameters in the direction that reduces loss.

Learning Rate

Definition

Learning rate controls the size of parameter updates during optimization.

Effects of Learning Rate

Learning Rate	Effect
Too Small	Slow training
Too Large	May skip optimal solution
Proper Value	Faster convergence

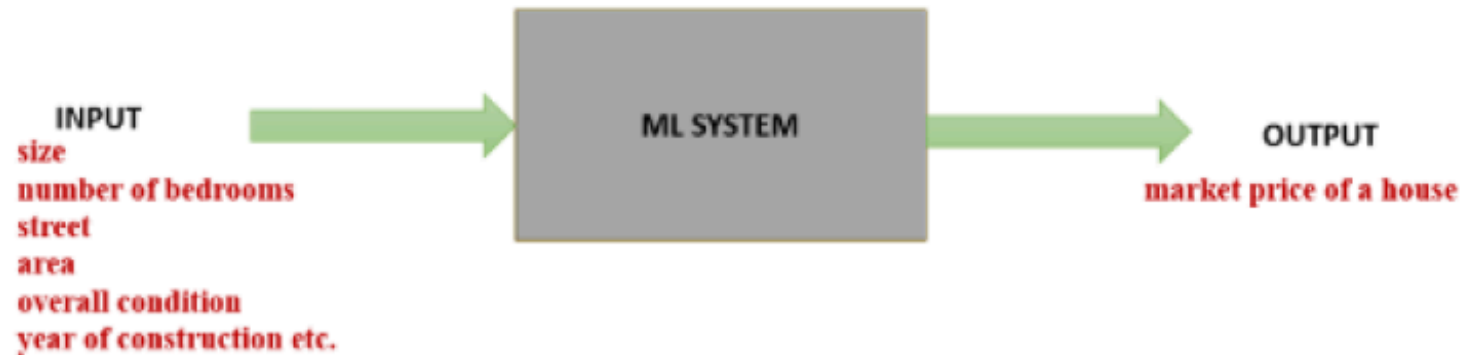
Relationship Between Loss Function and Optimization

Loss Function	Optimization
Measures error	Minimizes error
Defines objective	Achieves objective
Evaluates predictions	Updates parameters
Indicates performance	Improves performance

REGRESSION :

- Regression is a type of supervised learning used to **predict continuous values(real values) based on input features.**

- House price prediction



Regression Models

- A regression model establishes a mathematical function that illustrates the **relationship between one or more independent variables and a dependent or target variable.**
- Use these techniques widely when the **dependent and independent variables are linked in a linear or non-linear fashion**, and the **target variable has a set of continuous values.**

Different regression models

- The different regression models are defined based on type of functions used to represent the relation between the dependent variable y and the independent variables

1. Simple linear regression:

- There is only one continuous independent variable x and the assumed
- relation between the independent variable and the dependent variable y is
$$y = a + bx$$

2. Multiple regression:

- There are more than one independent variable, say $x_1, x_2, \dots, x_n$,
- the assumed relation between the independent variables and the dependent variable is
- $y = a_0 + a_1x_1 + \dots + a_nx_n$

3. Polynomial regression:

- There is only one continuous independent variable x and the assumed
- model is
- $y = a_0 + a_1x + \dots + a_nx^n$
- integer $n > 1$.

4. Logistic /Logit regression

- The dependent variable is binary, that is, a variable which takes only the
- values 0 and 1.
- The assumed model involves certain probability distributions

Linear regression with single independent variable

A simple example of a regression equation to predict the glucose level given the age.

SUBJECT	AGE X	GLUCOSE LEVEL Y
1	43	99
2	21	65
3	25	79
4	42	75
5	57	87
6	59	81
7	55	?

Estimated
(or predicted)
Y value for
observation i

Estimate of
the regression
intercept

Estimate of the
regression slope

Value of X for
observation i

$$\hat{Y}_i = b_0 + b_1 X_i$$

$$b_0 = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$b_1 = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

Linear regression with single independent variable

Step 1

SUBJECT	AGE X	GLUCOSE LEVEL Y	XY	X ²	Y ²
1	43	99	4257	1849	9801
2	21	65	1365	441	4225
3	25	79	1975	625	6241
4	42	75	3150	1764	5625
5	57	87	4959	3249	7569
6	59	81	4779	3481	6561
Σ	247	486	20485	11409	40022

Linear regression with single independent variable

Step 2
Find b_0 :

$$b_0 = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$b_0 = \frac{(486)(11409) - (247)(20485)}{6(11409) - (247)^2}$$

$$b_0 = \frac{4848979}{7445} = 65.14$$

Linear regression with single independent variable

Find b_1 :

$$b_1 = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

$$b_1 = \frac{6(20485) - (247)(486)}{6(11409) - (247)^2}$$

$$b_1 = \frac{2868}{7445} = 0.385335$$

- **Step 3:** Insert the values into the equation.

$$y' = b_0 + b_1x$$

$$y' = 65.14 + 0.385225x$$

- **Step 4:** Prediction – the value of y for the given value of $x = 55$

$$y' = 65.14 + (0.385225 * 55)$$

$$y' = 86.327$$

Hence, the glucose level for the given age 55 is 86.327



Multiple linear regression



芥 It is also known simply as multiple regression.

芥 It is a statistical technique that uses several explanatory variables to predict the outcome of a response variable.

芥 The goal of multiple linear regression is to model the linear relationship between the explanatory

(independent) variables and response
(dependent) variables.

Multiple linear regression

Idea: Examine the linear relationship between 1 dependent (Y) & 2 or more independent variables (X_i)

Multiple Regression Model with k Independent Variables:

The diagram shows the equation $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$. Above the equation, three labels in pink boxes are connected to their respective terms by arrows: 'Y-intercept' points to β_0 , 'Population slopes' points to β_1 and β_2 , and 'Random Error' points to ε .

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$

Labels and arrows:

- Y-intercept points to β_0
- Population slopes points to β_1 and β_2
- Random Error points to ε



Linear regression with multiple independent variables

Predict the value of Y given X_1 and X_2

SUBJECT	Y	X_1	X_2
1	-3.7	3	8
2	3.5	4	5
3	2.5	5	7
4	11.5	6	3
5	5.7	2	1
6	?	3	2

Linear Regression 2 independent variable

$$a = \bar{Y} - b_1 \bar{X}_1 - b_2 \bar{X}_2$$

$$b_1 = \frac{(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

$$b_2 = \frac{(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$


$$\sum x_1^2 = \sum X_1 X_1 - \frac{(\sum X_1)(\sum X_1)}{N}$$

$$\sum x_2^2 = \sum X_2 X_2 - \frac{(\sum X_2)(\sum X_2)}{N}$$

$$\sum x_1 y = \sum X_1 Y - \frac{(\sum X_1)(\sum Y)}{N}$$

$$\sum x_2 y = \sum X_2 Y - \frac{(\sum X_2)(\sum Y)}{N}$$

$$\sum x_1 x_2 = \sum X_1 X_2 - \frac{(\sum X_1)(\sum X_2)}{N}$$




SUBJECT	Y	X ₁	X ₂	X ₁ X ₁	X ₂ X ₂	X ₁ X ₂	X ₁ Y	X ₂ Y
1	-3.7	3	8	9	64	24	-11.1	-29.6
2	3.5	4	5	16	25	20	14	17.5
3	2.5	5	7	25	49	35	12.5	17.5
4	11.5	6	3	36	9	18	69	34.5
5	5.7	2	1	4	1	2	11.4	5.7
Σ	19.5	20	24	90	148	99	95.8	45.6


$$\sum x_1^2 = \sum X_1 X_1 - \frac{(\sum X_1)(\sum X_1)}{N} = 10$$

$$\sum x_2^2 = \sum X_2 X_2 - \frac{(\sum X_2)(\sum X_2)}{N} = 32.8$$

$$\sum x_1 y = \sum X_1 Y - \frac{(\sum X_1)(\sum Y)}{N} = 17.8$$

$$\sum x_2 y = \sum X_2 Y - \frac{(\sum X_2)(\sum Y)}{N} = -48$$

$$\sum x_1 x_2 = \sum X_1 X_2 - \frac{(\sum X_1)(\sum X_2)}{N} = 3$$

$$b_1 = \frac{(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2} = \frac{32.8 * 17.8 - 3 * (-48)}{10 * 32.8 - 3 * 3} = 2.28$$

$$b_2 = \frac{(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2} = \frac{10 * (-48) - 3 * 17.8}{10 * 32.8 - 3 * 3} = -1.67$$

$$a = \bar{Y} - b_1 \bar{X}_1 - b_2 \bar{X}_2 = \frac{19.5}{5} - \frac{2.28 * 20}{5} - \frac{-1.67 * 24}{5} = 2.796$$



Final Regression equation or Model is:

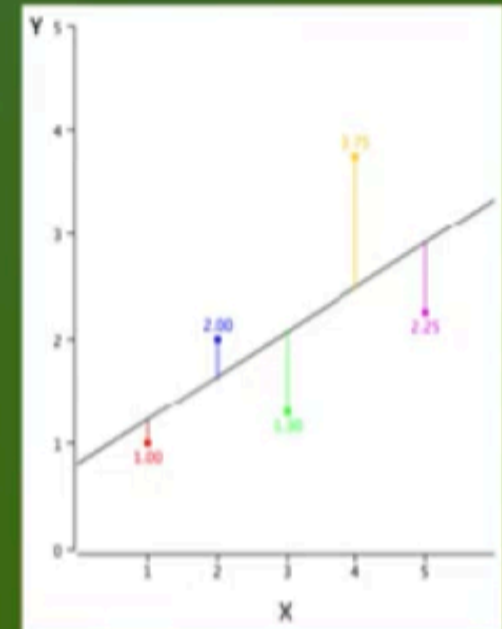
$$Y = 2.796 + 2.28 x_1 - 1.67 x_2$$

Now given $x_1 = 3$ *and* $x_2 = 2$ $Y = ?$

$$\begin{aligned} Y &= 2.796 + 2.28 * 3 - 1.67 * 2 \\ &= 6.296 \end{aligned}$$

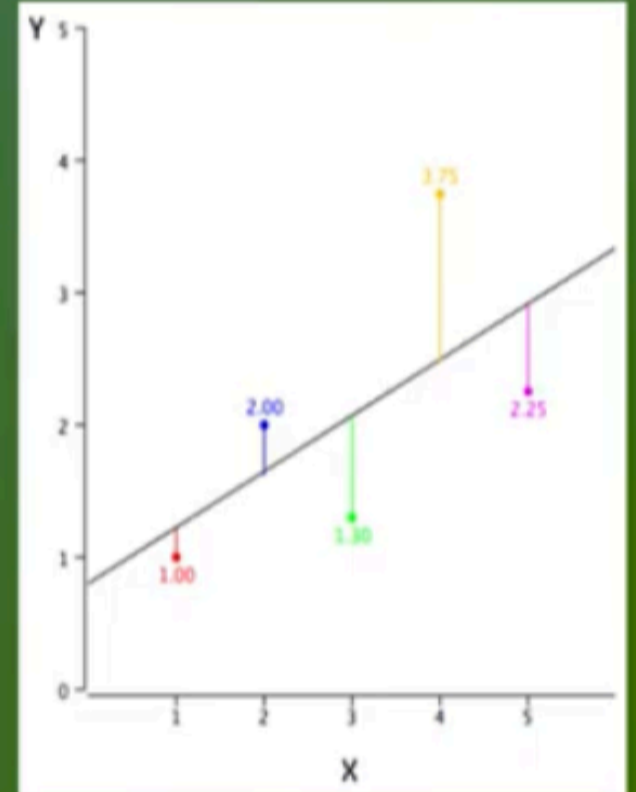
MATRIX METHOD FOR SIMPLE LINEAR REGRESSION

- Find the best fit line through a set of data points : $(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)$
- Best fit line
- Errors $\epsilon_i = y_i - (\beta_0 + \beta_1 x_i)$
- To minimize $\sum \epsilon_i^2$



MATRIX METHOD FOR SIMPLE LINEAR REGRESSION

- β_0 intercept and β slope
- $\beta_0 + X\vec{\beta} = \vec{y}$
- To simplify, add β_0 to $\vec{\beta}$
- Adding an extra column with 1's in X matrix and adding an extra variable in $\vec{\beta}$



MATRIX METHOD FOR SIMPLE LINEAR REGRESSION

$$\begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \times \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

- Least square matrix problem is :

$$\begin{bmatrix} \beta_0 + \beta_1 x_1 \\ \vdots \\ \beta_0 + \beta_1 x_n \end{bmatrix} \text{ is close to } \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

MATRIX METHOD FOR SIMPLE LINEAR REGRESSION

- Initial equation : $X\vec{\beta} = \vec{y}$
- Multiplying both sides by X^T : $X^T X \vec{\beta} = X^T \vec{y}$ where

$$X^T X = \begin{bmatrix} 1 & \dots & 1 \\ x_1 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} N & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$$

MATRIX METHOD FOR SIMPLE LINEAR REGRESSION

$$X^T \vec{y} = \begin{bmatrix} 1 & \dots & 1 \\ x_1 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \sum y \\ \sum X_i y_i \end{bmatrix}$$

$$\vec{\beta} = (X^T X)^{-1} X^T \vec{y}$$

MATRIX METHOD FOR SIMPLE LINEAR REGRESSION

$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} N & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum y \\ \sum x_i y_i \end{bmatrix}$$

Example:

A simple linear regression problem with 4 data points, (1,1), (2,3), (3,3) and (4,5)

$$X = [1 \ 2 \ 3 \ 4]$$

$$Y = [1 \ 3 \ 3 \ 5]$$

Convert in to matrix form,

$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 3 \\ 5 \end{bmatrix}$$

Here $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \quad y = \begin{bmatrix} 1 \\ 3 \\ 3 \\ 5 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} N & \sum X_i \\ \sum X_i & \sum X_i^2 \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}$$

$$X^T \vec{y} = \begin{bmatrix} \sum y \\ \sum x_i y_i \end{bmatrix} = \begin{bmatrix} 12 \\ 36 \end{bmatrix}$$

$$X^T X \vec{\beta} = X^T \vec{y}$$

$$\begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix} \times \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 12 \\ 36 \end{bmatrix}$$

$$4\beta_0 + 10\beta_1 = 12 \quad \text{--- ①}$$

$$10\beta_0 + 30\beta_1 = 36 \quad \text{--- ②}$$

$$\textcircled{1} \times 3$$

$$12\beta_0 + 30\beta_1 = 36$$

$$- \quad 10\beta_0 + 30\beta_1 = 36$$

$$2\beta_0 = 0$$

$$\beta_0 = 0$$

$$\therefore \beta_1 = 36/30 = 6/5 = 1.2$$

$$\therefore \text{the best line is } y = 0 + 6/5 X$$



MATRIX METHOD FOR MULTIPLE LINEAR REGRESSION

- 
- 1) Add a new column beginning with all 1's for intercept in X matrix
 - 2) Take transpose of X matrix
 - 3) Multiply X transpose and X matrices
 - 4) Find the inverse of this matrix
 - 5) Multiply X transpose with y matrix
 - 6) Multiply both the matrices to find intercept and the coefficient

Problem 2

Question: Fit a multiple linear regression model to the following data:

x_1	1	1	2	0
x_2	1	2	2	1
y	3.25	6.5	3.5	5.0



芥 The multiple linear regression model for this problem has the form

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2.$$

芥 The computations are

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 2 \\ 1 & 0 & 1 \end{bmatrix}, \quad Y = \begin{bmatrix} 3.25 \\ 6.5 \\ 3.5 \\ 5.0 \end{bmatrix}, \quad B = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$



□ The required model is

$$y = 2.0625 - 2.3750x_1 + 3.2500x_2.$$



Gradient descent algorithm

- Gradient Descent is an optimization algorithm used for minimizing the cost function in various machine learning algorithms. It is basically used for updating the parameters of the learning model.
- Gradient descent (GD) is an iterative first-order optimisation algorithm used to find a **local minimum/maximum** of a given function. This method is commonly used in machine learning (ML) and deep learning(DL) to minimise a cost/loss function (e.g. in a linear regression).

- Gradient is a slope of a curve at a given point in a specified direction.

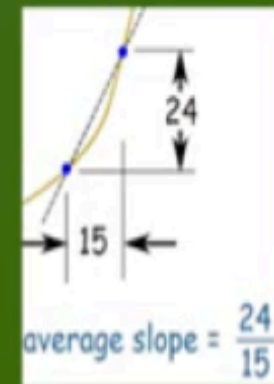
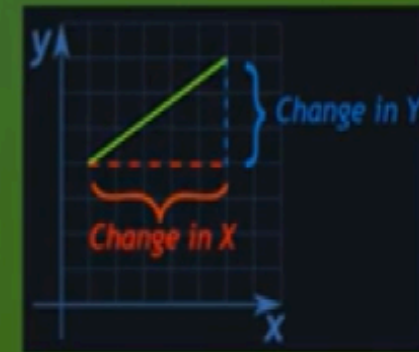
Gradient descent algorithm

- In the case of a **univariate** function, it is simply the **first derivative** at a selected point.
 - In the case of a **multivariate** function, it is a vector of derivatives in each main direction (along variable axes). Because we are interested only in a slope along one axis and we don't care about others. These derivatives are called **partial derivatives**.
- ✿ A maximum or minimum is said to be **local** if it is the largest or smallest value of the function, respectively, within **a given range**.

芥 However, a maximum or minimum is said to be *global* if it is the largest or smallest value of the function, respectively, on the **entire domain of a function**.

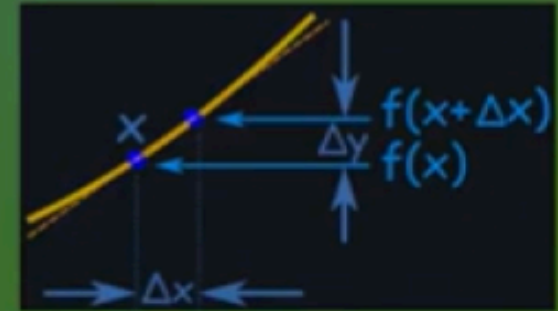
Gradient descent algorithm

- Slope - ratio of altitude change to horizontal distance between any two points on the line
- Gives the measure of its steepness and direction
- Average slope between two points
- To find slope at a point, we use derivative
- Uses slope formula



Gradient descent algorithm

- x changes from x to $x + \Delta x$
- y changes from $f(x)$ to $f(x + \Delta x)$
- For function x^2 , rate of change at any point is $2x$
- For multiple variables, partial derivative with respect to one of those variables with the others held constant
- Determine how changes in direction of a variable influence a function





Gradient descent algorithm

- 
- Gives the direction of maximum change
 - Magnitude indicates maximum rate of change
 - Points in direction of steepest increase in objective function
 - Function we want to minimize or maximize - objective function
 - When minimizing it, cost function, loss function or error function

Gradient descent procedure

- The procedure starts off with initial values for the coefficient or coefficients for the function.

Parameters/**coefficient** = **start with a small random value**

- The cost of the coefficients is evaluated by plugging them into the function and calculating the cost.

cost = evaluate(f(coefficient))

- Compute the gradient: The derivative of the cost is calculated. The derivative is a concept from calculus and refers to the slope of the function at a given point. We need to know the slope so

that we know the direction (sign) to move the coefficient values in order to get a lower cost on the next iteration.

delta = derivative(cost)

Gradient descent procedure

✦ Now that we know from the derivative which direction is downhill, we can now update the coefficient values.

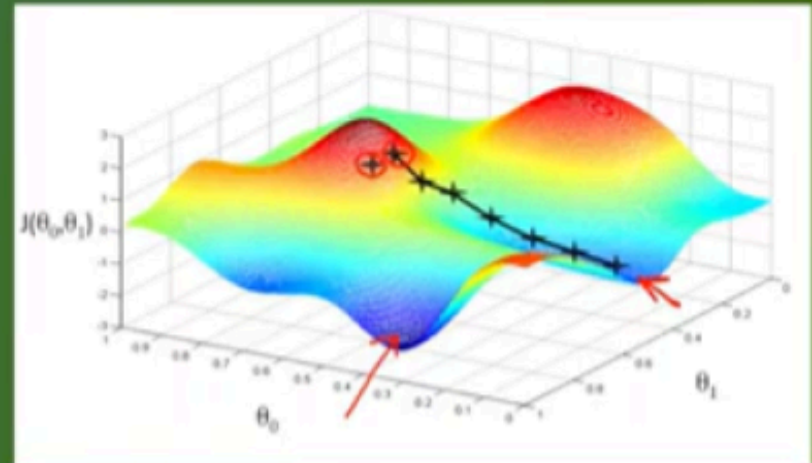
✦ A learning rate parameter (alpha) must be specified that controls how much the coefficients can change on each update.

$$\text{coefficient} = \text{coefficient} - (\text{alpha} * \text{delta})$$

✂ This process is repeated until the cost of the coefficients (cost) is 0.0 or close enough to zero to be good enough.

Gradient descent algorithm

- If we update variables or parameters of some cost function in the direction of the negative gradient in an iterative manner to reach the minimum of some cost function - gradient descent algorithm
- In context of linear regression
- $Y = mX + b$



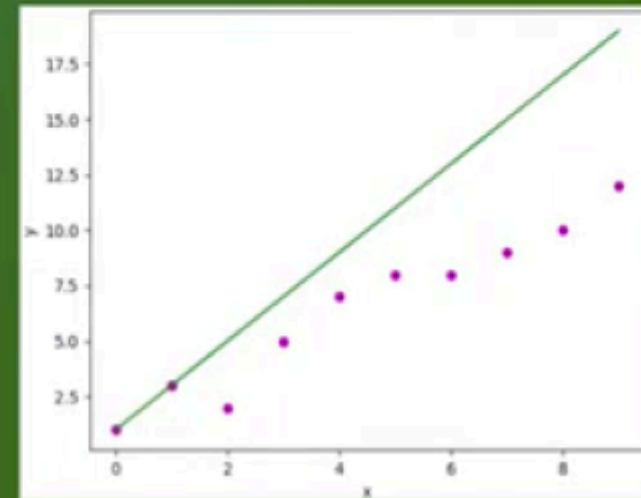
Gradient descent algorithm

1) Initialize coefficients m and b with random values :

- If $m = 1$ and $b = 2$, then equation becomes $y = 1 * x + 2$
- Not best fitted for historical data

2) Compute gradient :

- Use the Sum of Squared Errors (SSE)
as our loss/ cost function



Gradient descent algorithm

- Gradient of SSE is its partial derivative with respect to m and b

$$SSE = \frac{1}{2}(Y - \bar{Y})^2$$

$$\bar{Y} = mx + b$$

$$\frac{\partial}{\partial m} SSE = -(Y - \bar{Y})X$$

Gradient descent algorithm

$$\frac{\partial}{\partial b} SSE = -(Y - \bar{Y})$$

$$\nabla SSE = \begin{bmatrix} \frac{\partial}{\partial m} SSE \\ \frac{\partial}{\partial b} SSE \end{bmatrix} = \begin{bmatrix} -(Y - \bar{Y})X \\ -(Y - \bar{Y}) \end{bmatrix}$$

Gradient descent algorithm

3) Update coefficients in the direction of optimal m and b :

- Using gradient

$$m = m - r * \frac{\partial}{\partial m} SSE$$

$$b = b - r * \frac{\partial}{\partial b} SSE$$

- r – learning rate

Gradient descent algorithm

4) Use new m and b for prediction :

- Calculate SSE of each data point to find out the total SSE, which is a sum of squared errors of data points, divided of 2
- Total SSE indicates error rate of model prediction

$$\text{Total SSE} = \frac{1}{2} \sum_{i=1}^n (Y - \bar{Y})^2$$

Gradient descent algorithm

5) Repeat steps 2, 3 and 4 :

- Until optimal values for coefficients m and b are found that reduces SSE to a minimum value
- Enable model to predict Y with highest accuracy

